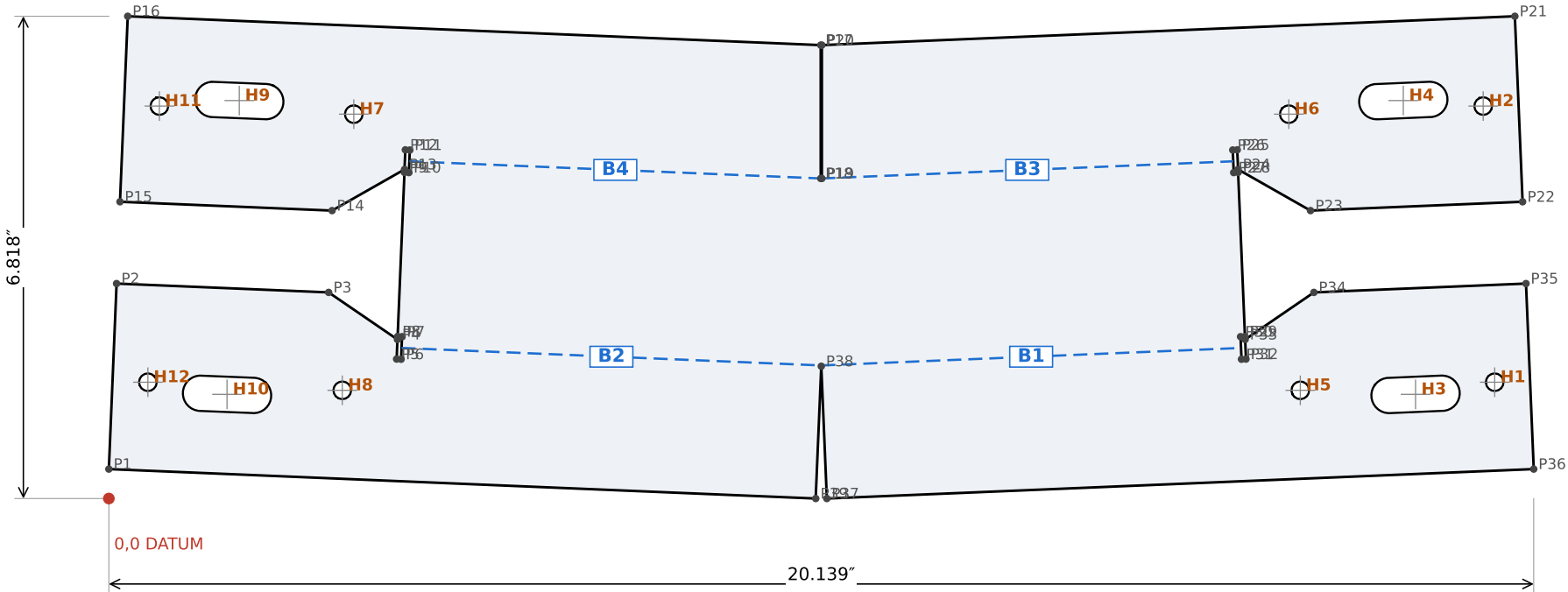


FF-1 — blank 1 of 1

Channel with ears — flat pattern, one piece (4 bends 90°, web on top) · blank 20.139" × 6.818" · HR steel sheet 10 ga (0.135") · 1 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.250"	X 19.588	Y 1.644
H2	Ø0.250"	X 19.425	Y 5.548
H3	slot 1.250 × 0.500"	@2° X 18.469	Y 1.473
H4	slot 1.250 × 0.500"	@2° X 18.296	Y 5.627
H5	Ø0.250"	X 16.840	Y 1.530
H6	Ø0.250"	X 16.678	Y 5.434
H7	Ø0.250"	X 3.462	Y 5.434
H8	Ø0.250"	X 3.299	Y 1.530
H9	slot 1.250 × 0.500"	@-2° X 1.844	Y 5.627
H10	slot 1.250 × 0.500"	@-2° X 1.670	Y 1.473
H11	Ø0.250"	X 0.714	Y 5.548
H12	Ø0.250"	X 0.552	Y 1.644

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1	(1)/(3) 90°	(10.070, 1.881) → (16.002, 2.129)
B2	(2)/(4) 90°	(4.137, 2.129) → (10.070, 1.881)
B3	radius-bend edge	(10.070, 4.525) → (15.892, 4.768)
B4	radius-bend edge	(4.247, 4.768) → (10.070, 4.525)
radius-bend edge = start/end of a large-radius bend (bump bend between them)		

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.416
P2	0.109	3.039
P3	3.107	2.914
P4	4.079	2.249
P5	4.067	1.972
P6	4.131	1.969
P7	4.144	2.288
P8	4.082	2.291
P9	4.178	4.610
P10	4.241	4.608
P11	4.254	4.927
P12	4.191	4.930
P13	4.179	4.653
P14	3.155	4.071
P15	0.158	4.196
P16	0.267	6.818
P17	10.062	6.410
P18	10.062	4.525
P19	10.078	4.525
P20	10.078	6.410
P21	19.873	6.818
P22	19.982	4.196
P23	16.985	4.071
P24	15.960	4.653
P25	15.949	4.930
P26	15.886	4.927
P27	15.899	4.608
P28	15.961	4.610
P29	16.058	2.291
P30	15.995	2.288
P31	16.009	1.969
P32	16.072	1.971
P33	16.061	2.249
P34	17.033	2.914
P35	20.030	3.039
P36	20.140	0.416
P37	10.148	0.000
P38	10.070	1.870
P39	9.991	0.000

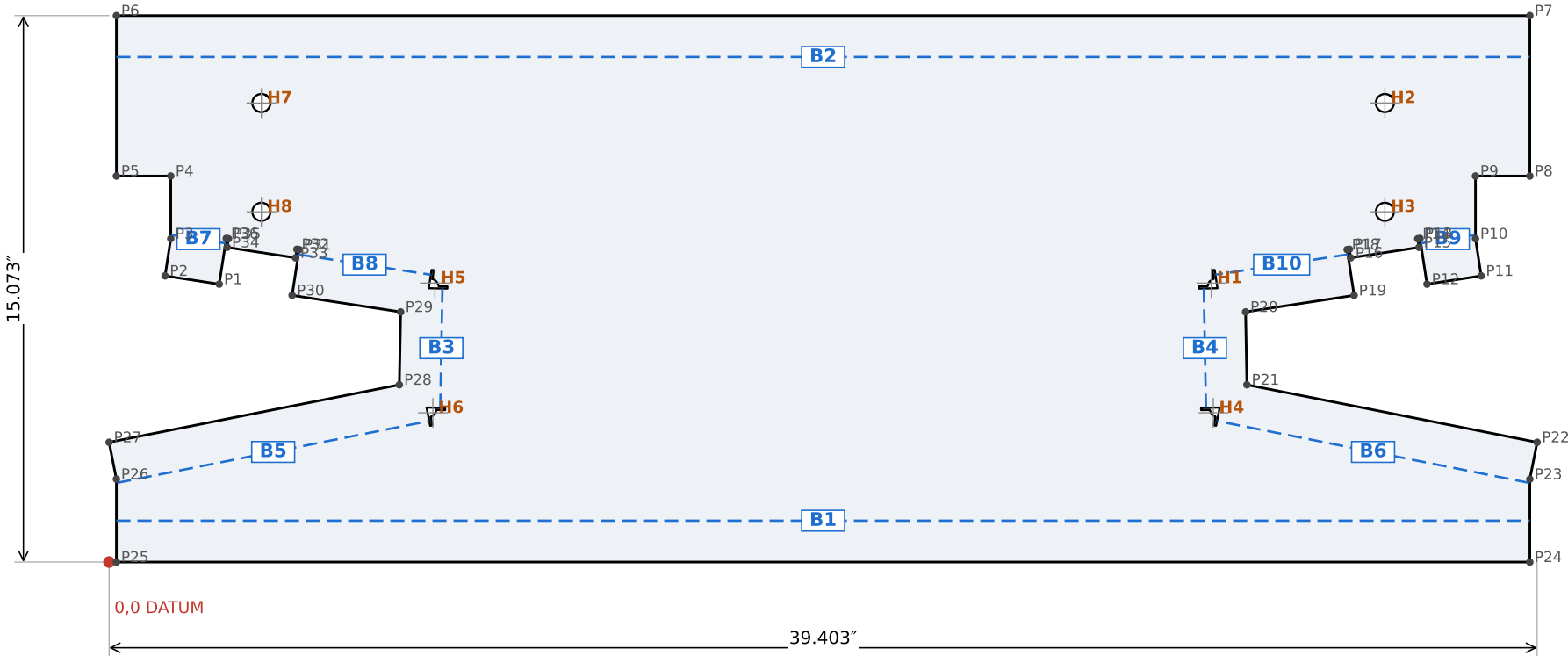
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
3. Flat pattern already includes bend allowance (K = 0.44, R = t).
4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * one blank: the web is a chevron 6"+6" (ends ¼" out); bend the walls 90° down on the dashed lines, each in 2 hits (6" to the middle); outer wall has a V-notch — closes after bending, weld it; inner wall has a slit — opens ~0.16" after bending, weld it; the ears reach past the bend line; 1/16" relief slit at the bend end

FF-1 — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel with ears — flat pattern, one piece (4 bends 90°, web on top)	QTY: 1 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FF-3 — blank 1 of 4

FF3-1 top plate — flat pattern, 10 bends 90° down · blank 39.403” × 15.073” · HR steel sheet 11 ga (0.120”) · 1 per vehicle



HOLES (centre X, Y from datum)

H1	slot	0.512 × 0.508"	@1°	X	30.413	Y	7.696
H2	Ø0.500"			X	35.202	Y	12.662
H3	Ø0.500"			X	35.202	Y	9.662
H4	slot	0.514 × 0.502"	@79°	X	30.469	Y	4.117
H5	slot	0.512 × 0.508"	@-1°	X	8.990	Y	7.696
H6	slot	0.514 × 0.502"	@-79°	X	8.934	Y	4.117
H7	Ø0.500"			X	4.202	Y	12.662
H8	Ø0.500"			X	4.202	Y	9.662

BENDS (bend centre line, from datum) IR = t = 11 ga (0.120")

B1	(3)	90° ↓	(0.202, 1.141) → (39.202, 1.141)
B2	(4)	90° ↓	(0.202, 13.933) → (39.202, 13.933)
B3	(5)	90° ↓	(9.138, 4.256) → (9.197, 7.546)
B4	(6)	90° ↓	(30.265, 4.256) → (30.206, 7.546)
B5	(7)	90° ↓	(0.222, 2.185) → (8.833, 3.892)
B6	(8)	90° ↓	(39.181, 2.185) → (30.570, 3.892)
B7	(9)	90° ↓	(1.717, 9.022) → (3.212, 8.793)
B8	(10)	90° ↓	(5.223, 8.486) → (8.878, 7.926)
B9	(11)	90° ↓	(37.686, 9.022) → (36.191, 8.793)
B10	(12)	90° ↓	(34.180, 8.486) → (30.526, 7.926)

OUTLINE POINTS (X, Y from datum)

P1	3.039	7.666
P2	1.545	7.895
P3	1.702	8.920
P4	1.702	10.649
P5	0.202	10.649
P6	0.202	15.073
P7	39.202	15.073
P8	39.202	10.649
P9	37.702	10.649
P10	37.702	8.920
P11	37.859	7.895
P12	36.364	7.666
P13	36.171	8.927
P14	36.109	8.918
P15	36.145	8.681
P16	34.258	8.392
P17	34.222	8.629
P18	34.160	8.619
P19	34.353	7.358
P20	31.358	6.899
P21	31.395	4.891
P22	39.403	3.304
P23	39.202	2.287
P24	39.202	0.000
P25	0.202	0.000
P26	0.202	2.287
P27	0.000	3.304
P28	8.008	4.891
P29	8.045	6.899
P30	5.050	7.358
P31	5.243	8.619
P32	5.181	8.629
P33	5.145	8.392
P34	3.258	8.681
P35	3.294	8.918
P36	3.232	8.927

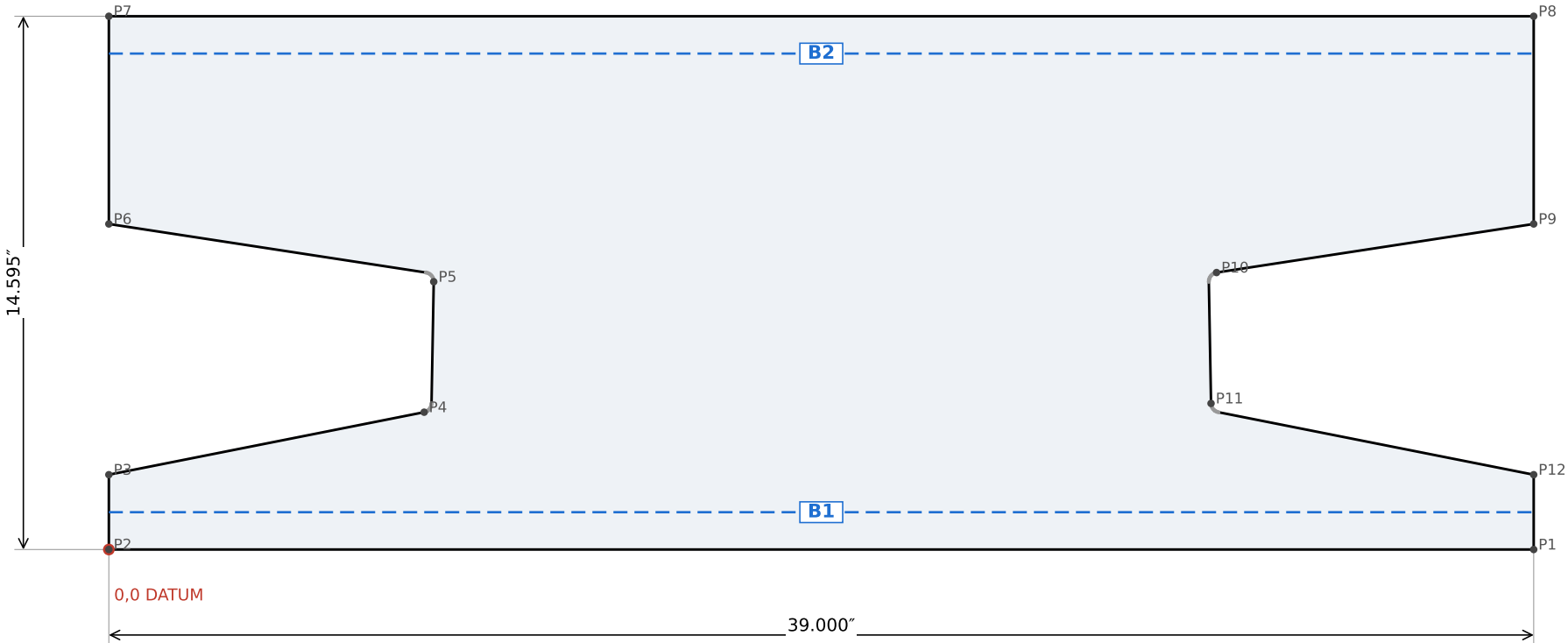
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 2. Material: HR steel sheet 11 ga (0.120") (t). Inside bend radius = t.
 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims 1½" bent down on the long sides and around the whole notch; leg ends open; in the 4 inner notch corners the rims are mitred — a gap stays in the corner; 1/16" reliefs at the bend ends; the 4 Ø½ holes are cut right away

FF-3 — blank 1 of 4	HR steel sheet 11 ga (0.120")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-1 top plate — flat pattern, 10 bends 90° down	QTY: 1 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FF-3 — blank 2 of 4

FF3-2 lower plate — flat pattern, 2 bends 90° up · blank 39.000″ × 14.595″ · HR steel sheet 11 ga (0.120″) · 1 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 11 ga (0.120″) (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * rims on the long sides bent up 1″ (to the FF3-1 underside), set in by t inside the FF3-1 rims; 1/16″ reliefs at the bend ends

HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 11 ga (0.120″)

B1 (1) 90° (0.000, 1.021) → (39.000, 1.021)

B2 (2) 90° (0.000, 13.574) → (39.000, 13.574)

OUTLINE POINTS (X, Y from datum)

P1	39.000	0.000
P2	0.000	0.000
P3	0.000	2.047
P4	8.630	3.758
P5	8.892	7.329
P6	0.000	8.910
P7	0.000	14.595
P8	39.000	14.595
P9	39.000	8.910
P10	30.320	7.581
P11	30.168	3.999
P12	39.000	2.047

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

8.677	3.772
8.720	3.795
8.739	3.809
8.774	3.843
8.801	3.883
8.820	3.928
8.830	3.975
8.832	3.999

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

8.889	7.377
8.883	7.400
8.866	7.445
8.841	7.485
8.808	7.520
8.770	7.549
8.726	7.569
8.680	7.581

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

30.273	7.569
30.230	7.549
30.192	7.520
30.159	7.485
30.134	7.445
30.117	7.400
30.111	7.377
30.108	7.329

FREE-FORM CURVE 4 (grey dots, follow DXF) — points X, Y

30.170	3.975
30.180	3.928
30.199	3.883
30.226	3.843
30.261	3.809
30.280	3.795
30.323	3.772
30.369	3.758

FF-3 — blank 2 of 4	HR steel sheet 11 ga (0.120″)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FF3-2 lower plate — flat pattern, 2 bends 90° up	QTY: 1 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 2 / 4

FF-3 — blank 3 of 4

Passage wall 1⅛ × 4.69" (instead of the tube) · blank 4.687" × 1.125" · HR steel sheet 11 ga (0.120") · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	4.687	0.000
P2	4.687	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 11 ga (0.120") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 3 of 4	HR steel sheet 11 ga (0.120")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅛ × 4.69" (instead of the tube)	QTY: 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FF-3 — blank 4 of 4

Passage wall 1⅞ × 5.01″ (instead of the tube) · blank 5.013″ × 1.125″ · HR steel sheet 11 ga (0.120″) · 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	5.013	0.000
P2	5.013	1.125
P3	0.000	1.125
P4	0.000	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 11 ga (0.120″) (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- * stands on FF3-2 under FF3-1, on the old tube wall line; weld top and bottom

FF-3 — blank 4 of 4	HR steel sheet 11 ga (0.120″)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Passage wall 1⅞ × 5.01″ (instead of the tube)	QTY: 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 4 / 4

FF-4 — cut list (tubes)

• Square tube 1 ½” — cut 39”

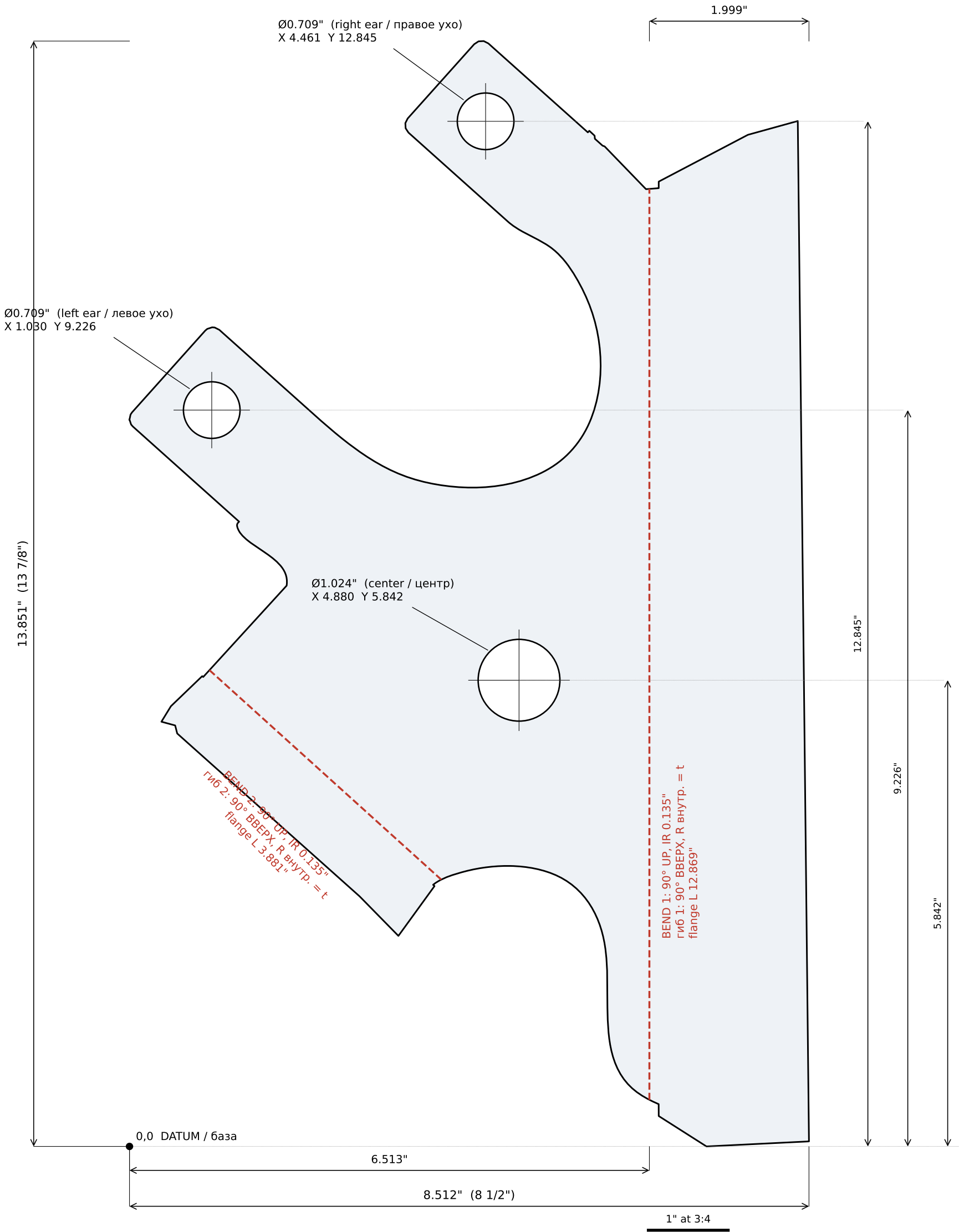
Stock: Square tube 1 ½” × 1 ½”, wall 3/16” · cut length 39.000” · QTY 1 per vehicle
wall 3/16”

Cut square, ±1/32”. Deburr. Lengths in inches.

FF-4 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Square tube 1 ½” — cut 39”	QTY: 1 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-1 — FLAT PATTERN / РАЗВЕРТКА (scale 3:4 / масштаб 3:4)

blank 8.512" × 13.851" · 10 ga (0.135") plate · 2 bends 90° up · IR 0.135"



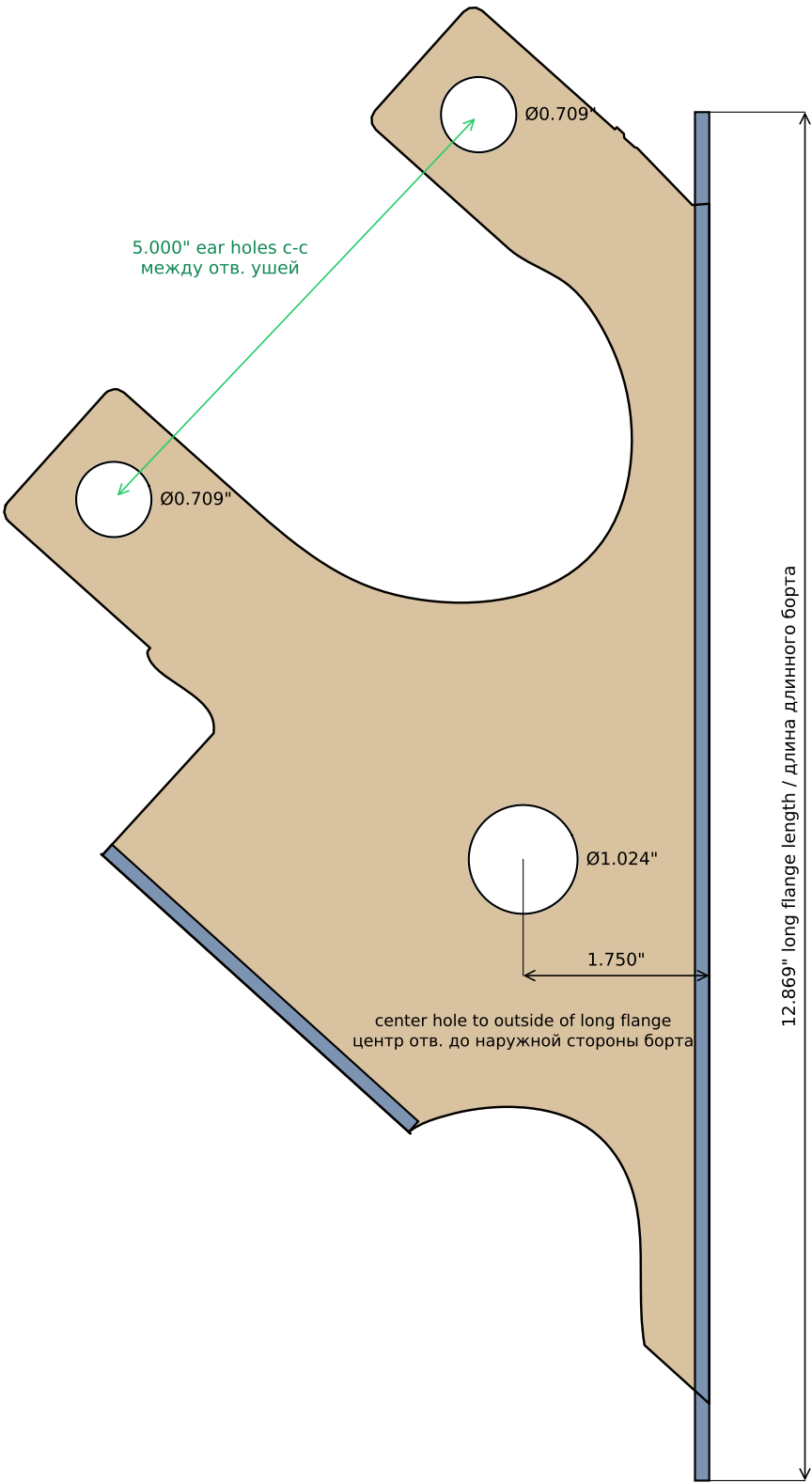
NOTES / ПРИМЕЧАНИЯ

- 1. Cut from the DXF file FR-1_flat_10ga.dxf (1:1, inches). This sheet is for checking. Резать по DXF (1:1, дюймы); чертёж — для проверки.
- 2. Both bends 90° UP to the SAME side (toward the viewer), inside radius 0.135" (R = t). Оба гiba 90° в одну сторону (к себе), R внутр. = t.
- 3. Flat pattern computed with K = 0.44, bend allowance 0.304", bend deduction 0.234" per bend.
If your tooling/K differs, keep the FINISHED sizes on sheet 2 and adjust the blank. Если ваш инструмент другой — держите готовые размеры (лист 2).
- 4. Red dashed lines = bend CENTER lines. Красный пунктир = средняя линия гiba.
- 5. Holes are laser cut in the flat (no drilling). Отверстия режутся лазером сразу.
- 6. QTY 2: FR-1 as shown + FL-1 = SAME blank flipped over (bends to the other side = mirror part).
2 шт: FR-1 как на чертеже + FL-1 = та же заготовка, перевернуть (гнуть в другую сторону = зеркальная деталь).
- 7. Deburr all edges. Снять заусенцы. Curved edges: smooth arcs/splines from the DXF.

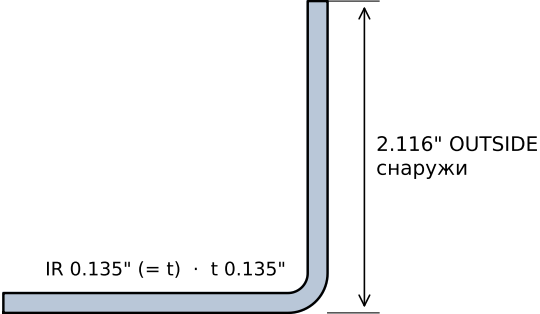
FR-1 · side plate / боковая пластина	A36 HR steel 10 ga (0.135")	inch / дюймы	Sienna WAV · 2026-10-02
FLAT PATTERN 3:4 / РАЗВЕРТКА 3:4	QTY 2 (FR-1 + FL-1 mirror)	cut ±0.010" · bend ±1°	SHEET 1 / 2

FR-1 — FINISHED PART / ГОТОВАЯ ДЕТАЛЬ

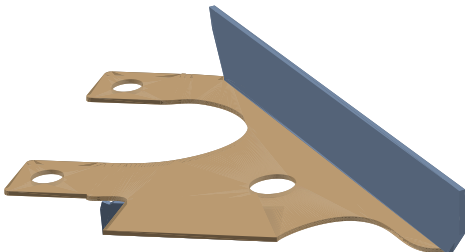
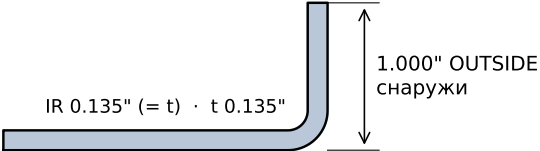
TOP VIEW (flanges up, toward viewer) / ВИД СВЕРХУ (борта вверх, к себе)



BEND 1 · long flange / длинный борт



BEND 2 · slant flange / косой борт



ISOMETRIC / ИЗОМЕТРИЯ

BEND TABLE / ТАБЛИЦА ГИБОВ

#	edge / кромка	angle	dir / направл.	IR	flange length	finished outside
1	long edge / длинная	90°	UP / вверх	0.135"	12.869"	2.116"
2	slant edge / косая	90°	UP / вверх	0.135"	3.881"	1.000"

Order: any (bends do not touch). Порядок гибов любой – они не пересекаются.
Both flanges on the SAME side. Оба борта в одну сторону.
FL-1 (mirror): same blank, bend both flanges to the OTHER side. FL-1: та же заготовка, гнуть в другую сторону.

- CHECK AFTER BENDING / ПРОВЕРИТЬ ПОСЛЕ ГИБКИ:
- long flange outside height 2.116" (±1/32")

• slant flange outside height 1.000" (±1/32")

• center hole Ø1.024" (26 mm) is 1.750" from the outside of the long flange

• ear holes Ø0.709" (18 mm) both, are 5.000" apart (c-c) – they bolt to FR-2

FR-1 · side plate / боковая пластина

A36 HR steel 10 ga (0.135")

inch / дюймы

Sienna WAV · 2026-10-02

FINISHED PART, BEND TABLE / ГОТОВАЯ ДЕТАЛЬ, ГИБЫ

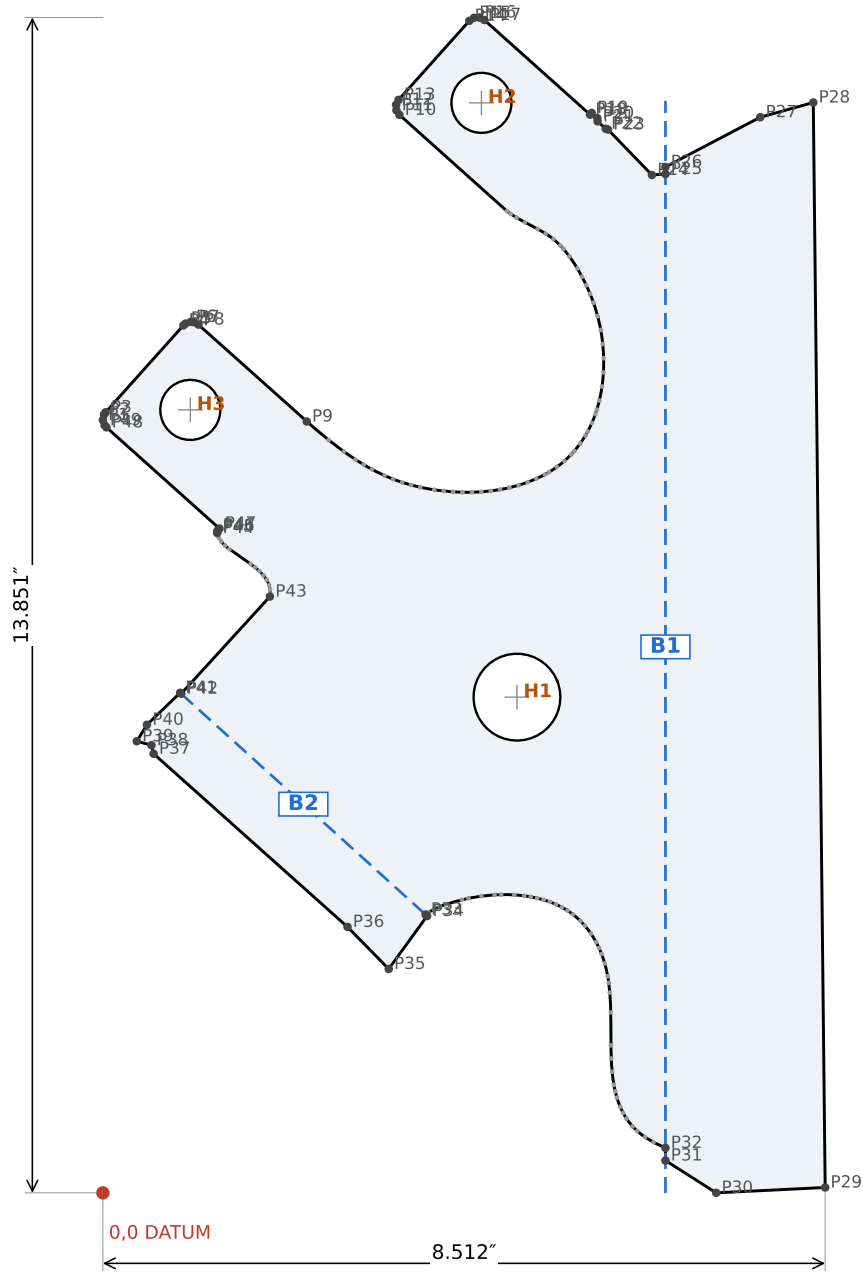
QTY 2 (FR-1 + FL-1 mirror)

cut ±0.010" · bend ±1°

SHEET 2 / 2

FR-1 — blank 1 of 1

Plate with ears — flat pattern (2 bends 90°) · blank 8.512″ × 13.851″ · HR steel sheet 10 ga (0.135″) · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135″) (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend 90° up on dashed lines

HOLES (centre X, Y from datum)

H1	Ø1.024″	X 4.880	Y 5.842
H2	Ø0.709″	X 4.461	Y 12.845
H3	Ø0.709″	X 1.030	Y 9.226

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135″)

B1	(1) 90°	(6.630, -0.001) → (6.630, 12.869)
B2	(2) 90°	(0.922, 5.882) → (3.803, 3.278)

OUTLINE POINTS (X, Y from datum)

P1	0.000	9.108
P2	0.013	9.171
P3	0.032	9.198
P4	0.949	10.222
P5	0.974	10.244
P6	1.035	10.264
P7	1.068	10.261
P8	1.125	10.232
P9	2.402	9.090
P10	3.495	12.704
P11	3.459	12.758
P12	3.456	12.823
P13	3.485	12.880
P14	4.319	13.812
P15	4.373	13.847
P16	4.437	13.851
P17	4.495	13.822
P18	5.741	12.706
P19	5.760	12.727
P20	5.827	12.665
P21	5.830	12.627
P22	5.927	12.540
P23	5.952	12.531
P24	6.470	11.995
P25	6.629	12.007
P26	6.630	12.089
P27	7.747	12.676
P28	8.372	12.849
P29	8.512	0.063
P30	7.228	0.000
P31	6.630	0.381
P32	6.630	0.530
P33	3.803	3.280
P34	3.821	3.262
P35	3.369	2.638
P36	2.883	3.134
P37	0.598	5.174
P38	0.573	5.275
P39	0.400	5.322
P40	0.519	5.516
P41	0.911	5.893
P42	0.924	5.882
P43	1.969	7.027
P44	1.347	7.777
P45	1.348	7.793
P46	1.354	7.808
P47	1.374	7.829
P48	0.041	9.021
P49	0.020	9.046

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.661	8.872
2.840	8.736
2.978	8.640
3.073	8.581

FR-1 — blank 1 of 1	HR steel sheet 10 ga (0.135″)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 1 / 1

3.170	8.527
3.318	8.453
3.421	8.410
3.525	8.373
3.632	8.341
3.740	8.315
3.905	8.285
4.016	8.271
4.127	8.261
4.238	8.256
4.350	8.256
4.461	8.261
4.572	8.271
4.682	8.288
4.791	8.310
4.899	8.339
5.005	8.374
5.109	8.416
5.209	8.465
5.258	8.493
5.351	8.554
5.438	8.624
5.519	8.701
5.556	8.743
5.626	8.830
5.687	8.924
5.715	8.972
5.763	9.073
5.804	9.177
5.822	9.230
5.851	9.338

5.883	9.502
5.899	9.668
5.902	9.835
5.891	10.003
5.868	10.168
5.831	10.331
5.782	10.490
5.742	10.594
5.696	10.695
5.646	10.796
5.591	10.894
5.498	11.036
5.428	11.123
5.391	11.164
5.309	11.239
5.220	11.303
5.123	11.359
4.973	11.436
4.876	11.491
4.806	11.536
4.762	11.570

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

6.490	0.596
6.389	0.657
6.297	0.728
6.242	0.783
6.171	0.875
6.131	0.942
6.097	1.012
6.057	1.122
6.037	1.197

FR-1 — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

6.022	1.273
6.001	1.427
5.991	1.583
5.987	1.738
5.985	2.127
5.974	2.360
5.962	2.475
5.950	2.552
5.915	2.704
5.877	2.814
5.847	2.886
5.812	2.955
5.752	3.055
5.681	3.148
5.628	3.204
5.540	3.281
5.476	3.326
5.409	3.366
5.340	3.400
5.267	3.429
5.193	3.453
5.118	3.473
5.042	3.489
4.965	3.501

4.849	3.511
4.694	3.515
4.616	3.511
4.461	3.497
4.308	3.471
4.232	3.454
4.045	3.400
3.972	3.374
3.901	3.341
3.834	3.302

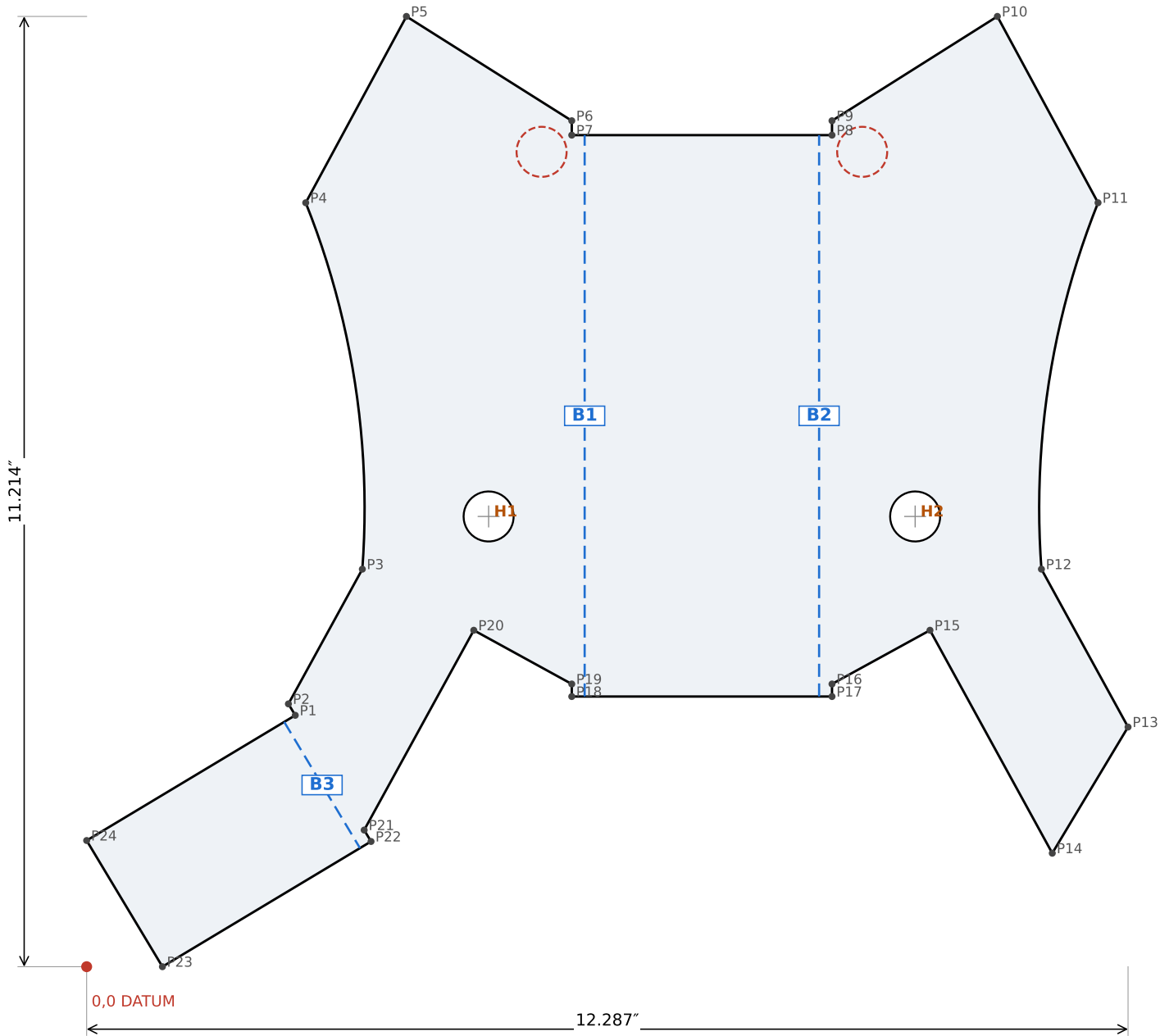
FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

1.972	7.061
1.970	7.111
1.959	7.161
1.931	7.222
1.892	7.277
1.846	7.326
1.795	7.370
1.741	7.411
1.491	7.585
1.440	7.628
1.394	7.678
1.367	7.720
1.353	7.752

FR-1 — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate with ears — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-10 (bent version) — blank 1 of 2

Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends) · blank 12.287" × 11.214" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591" X 4.743 Y 5.313
H2 Ø0.591" X 9.775 Y 5.313

DRILL AFTER WELDING (red dashed — do NOT laser cut)

D1 Ø0.591" X 5.368 Y 9.616
D2 Ø0.591" X 9.150 Y 9.616

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1 (2) 90° (5.876, 3.188) → (5.876, 9.813)
B2 (3) 90° (8.642, 3.188) → (8.642, 9.813)
B3 (1) 90° (2.331, 2.888) → (3.225, 1.399)

CURVED EDGES (radius, centre X, Y)

R1 R 9.700" centre -6.421, 5.410
R2 R 9.701" centre 20.940, 5.410

OUTLINE POINTS (X, Y from datum)

P1 2.461 2.967
P2 2.380 3.101
P3 3.253 4.691
P4 2.585 9.015
P5 3.773 11.214
P6 5.724 9.983
P7 5.724 9.813
P8 8.794 9.813
P9 8.794 9.983
P10 10.745 11.214
P11 11.933 9.015
P12 11.265 4.691
P13 12.287 2.828
P14 11.393 1.339
P15 9.950 3.970
P16 8.794 3.336
P17 8.794 3.188
P18 5.723 3.188
P19 5.723 3.336
P20 4.568 3.970
P21 3.274 1.612
P22 3.355 1.478
P23 0.894 0.000
P24 0.000 1.489

NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bends (1)(2)(3) 90° up, R = t; the 2 Ø0.591 holes near A (½" from A-B, ⅝" from A-H) are not in the flat — drill after bending

FR-10 (bent version) — blank 1 of 2	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Part 1 "U-body" — 2 sides + wall A-H + flange E-F (3 bends)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 3

FR-10 (bent version) — blank 2 of 2

Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends) · blank 12.478" × 2.716" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

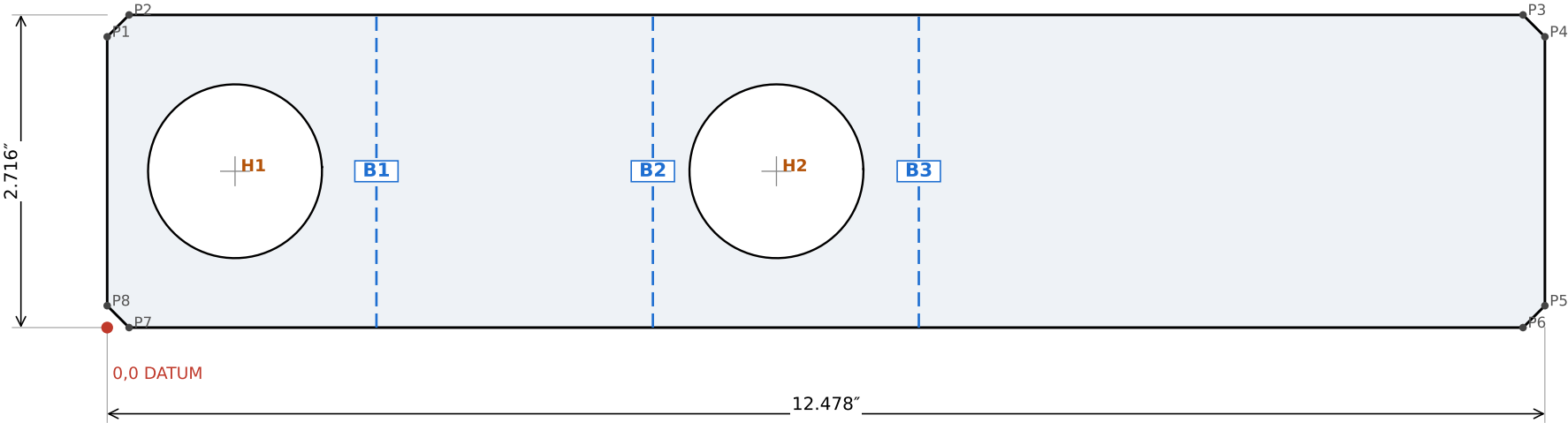
H1	Ø1.510"	X 1.110	Y 1.358
H2	Ø1.510"	X 5.809	Y 1.358

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1	(5) B 93.9°	(2.337, 0.000) → (2.337, 2.716)
B2	(6) C 86.1°	(4.736, 0.000) → (4.736, 2.716)
B3	(4) X 79.2° ↓	(7.044, 0.000) → (7.044, 2.716)

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.526
P2	0.190	2.716
P3	12.288	2.716
P4	12.478	2.526
P5	12.478	0.190
P6	12.288	0.000
P7	0.190	0.000
P8	0.000	0.190



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * (4) X 79.2° opposite way; (5) B 93.9°, (6) C 86.1° up; corners 3/16"×45°

FR-10 (bent version) — blank 2 of 2	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Part 2 "strip" — A-B · B-C · shelf · inner plate (3 bends)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-10 (bent version) — cut list (tubes)

- **Tube OD 1.5 / ID 1.181 — cut 3.42” (through A-B and the shelf)**

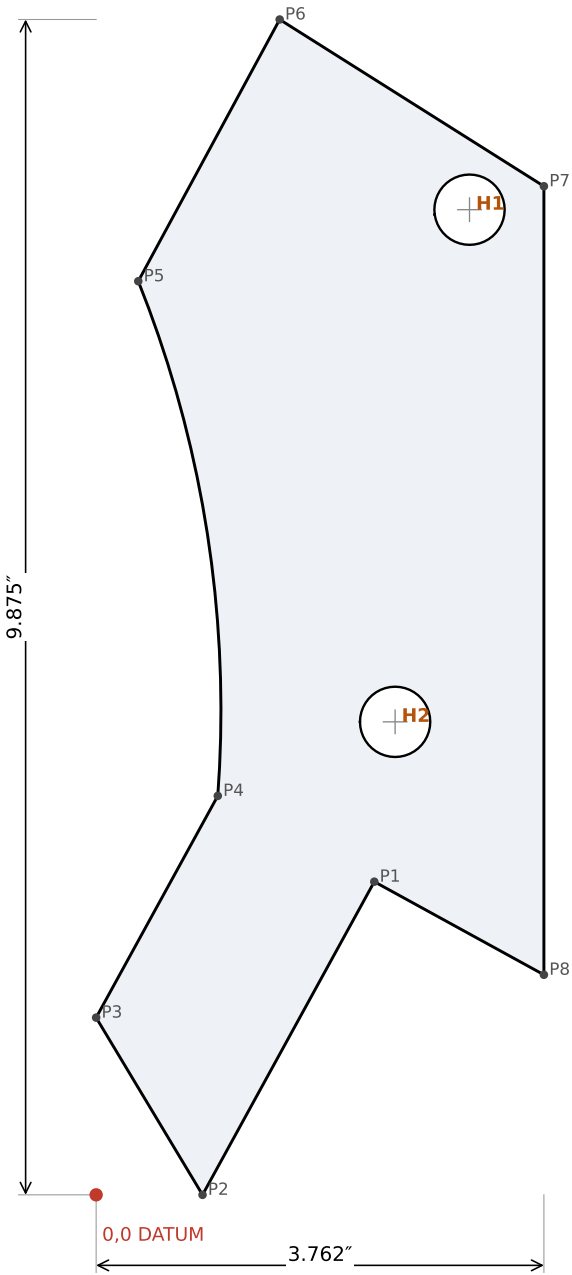
Stock: DOM round tube · cut length 3.420” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-10 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube OD 1.5 / ID 1.181 — cut 3.42” (through A-B and the shelf)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 3 / 3

FR-10 (welded version) — blank 1 of 7

Side plate (right + left) · blank 3.762” × 9.875” · HR steel sheet 10 ga (0.135”) · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.591"	X 3.137	Y 8.277
H2	Ø0.591"	X 2.512	Y 3.974

CURVED EDGES (radius, centre X, Y)

R1	R 9.700"	centre -8.651, 4.071
----	----------	----------------------

OUTLINE POINTS (X, Y from datum)

P1	2.337	2.631
P2	0.894	0.000
P3	0.000	1.489
P4	1.022	3.352
P5	0.354	7.676
P6	1.542	9.875
P7	3.762	8.474
P8	3.762	1.849

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 1 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate (right + left)	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 7

FR-10 (welded version) — blank 2 of 7

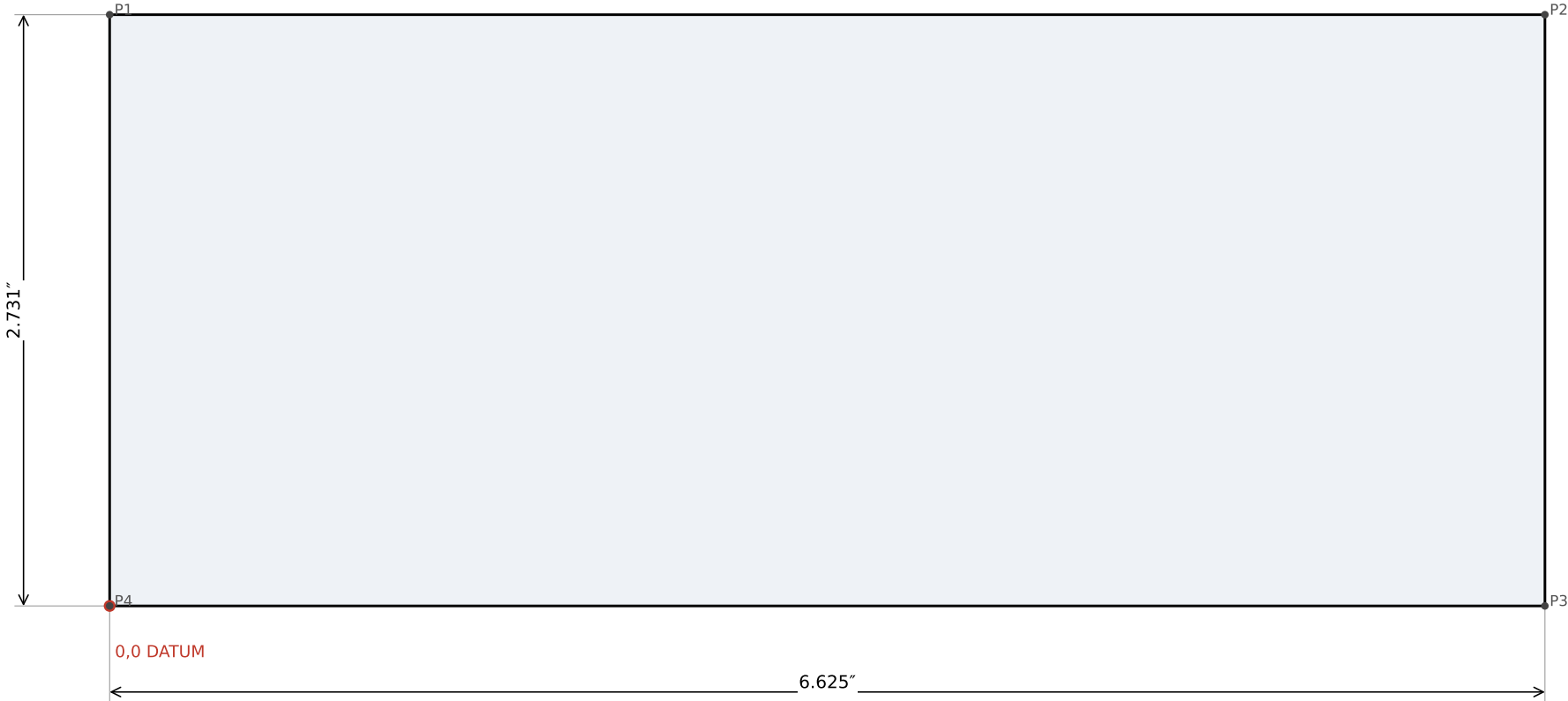
Plate A-H · blank 6.625" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.731
P2	6.625	2.731
P3	6.625	0.000
P4	0.000	0.000



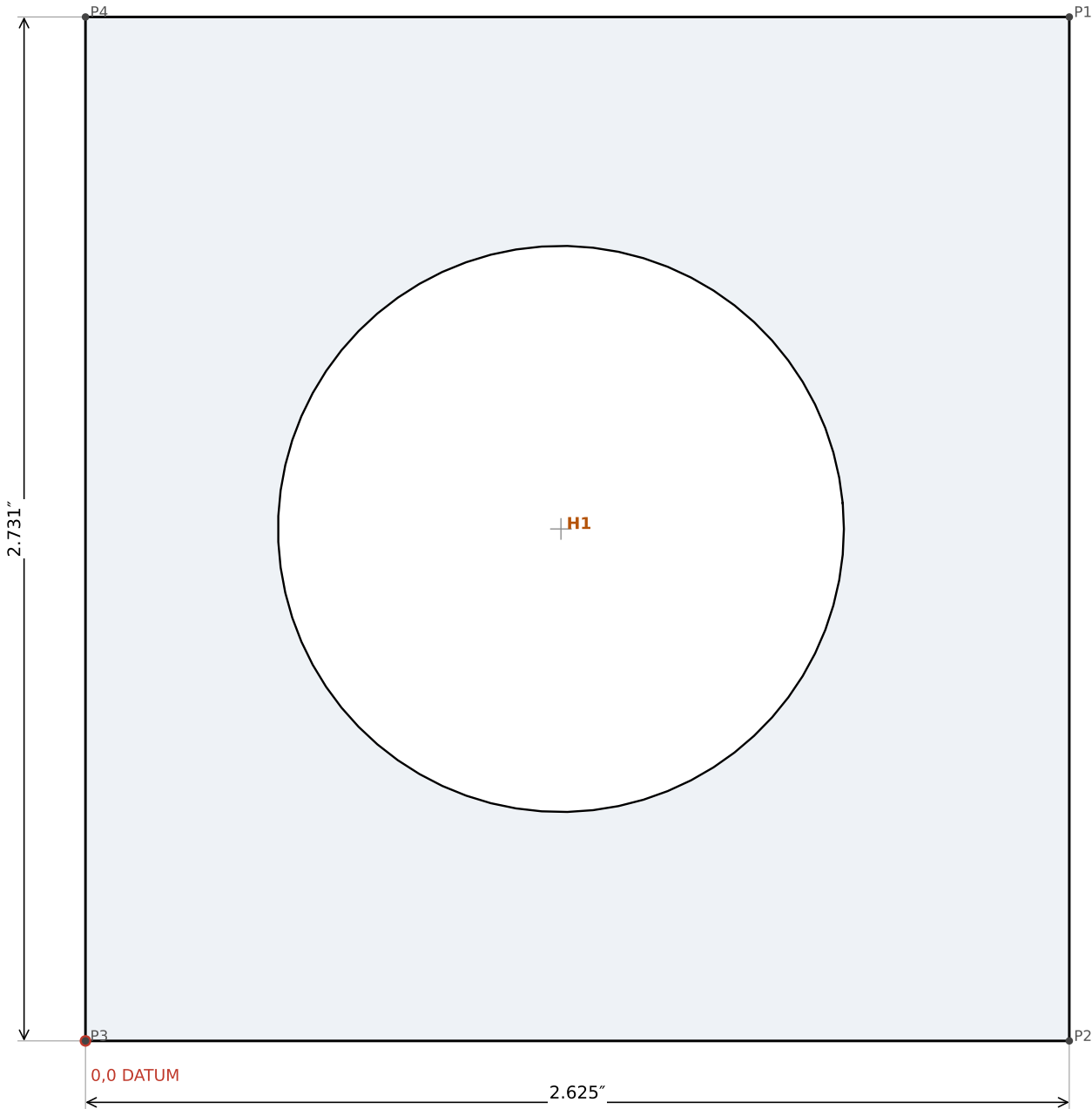
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 2 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate A-H	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 7

FR-10 (welded version) — blank 3 of 7

Plate A-B (hole Ø1.5 for the tube) · blank 2.625” × 2.731” · HR steel sheet 10 ga (0.135”) · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.510” X 1.269 Y 1.366

OUTLINE POINTS (X, Y from datum)

P1 2.625 2.731
P2 2.625 0.000
P3 0.000 0.000
P4 0.000 2.731

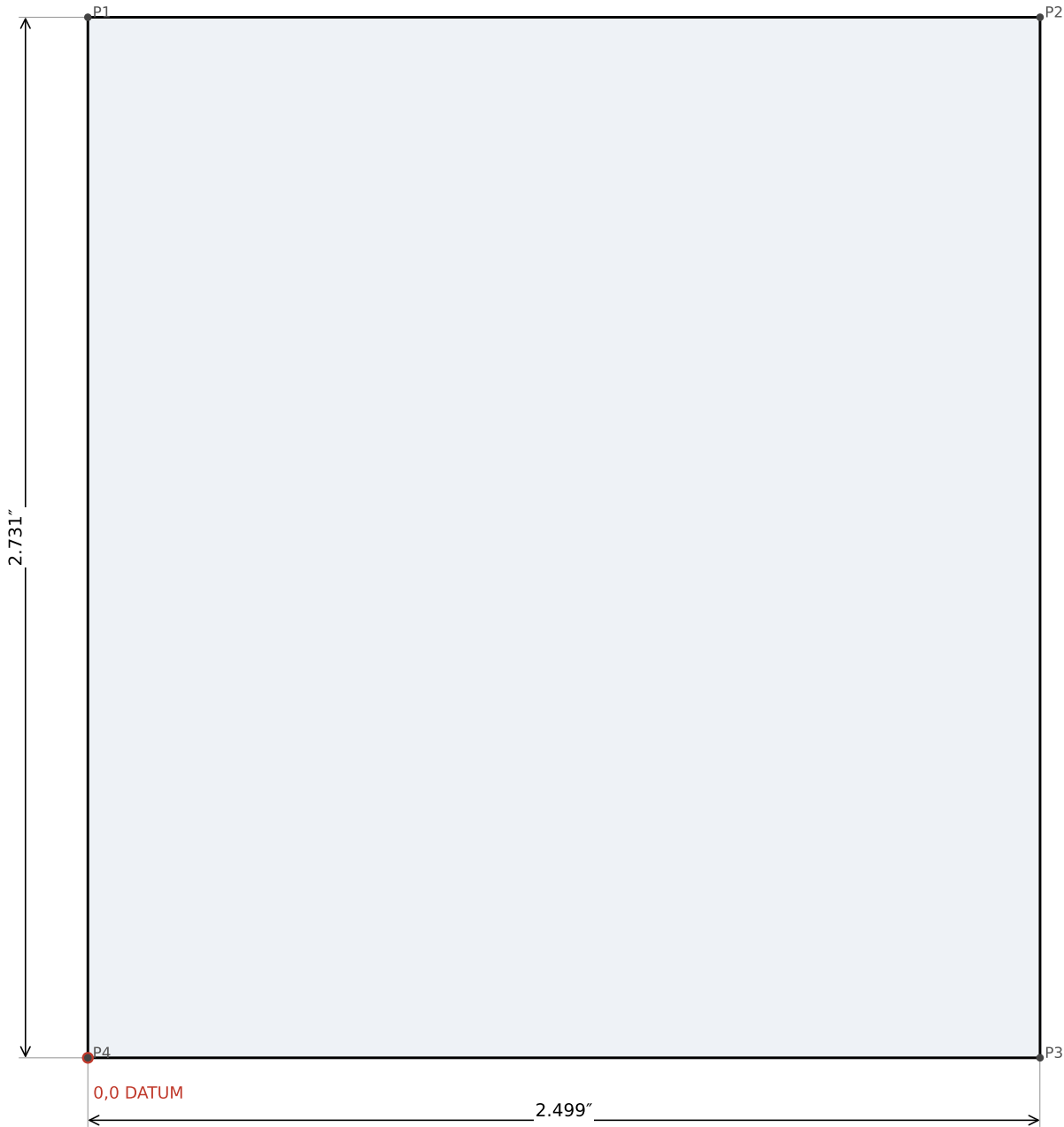
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135”) (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * tube OD 1.5 / ID 1.181 passes through

FR-10 (welded version) — blank 3 of 7	HR steel sheet 10 ga (0.135”)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate A-B (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 3 / 7

FR-10 (welded version) — blank 4 of 7

Plate B-C · blank 2.499" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.731
P2	2.499	2.731
P3	2.499	0.000
P4	0.000	0.000

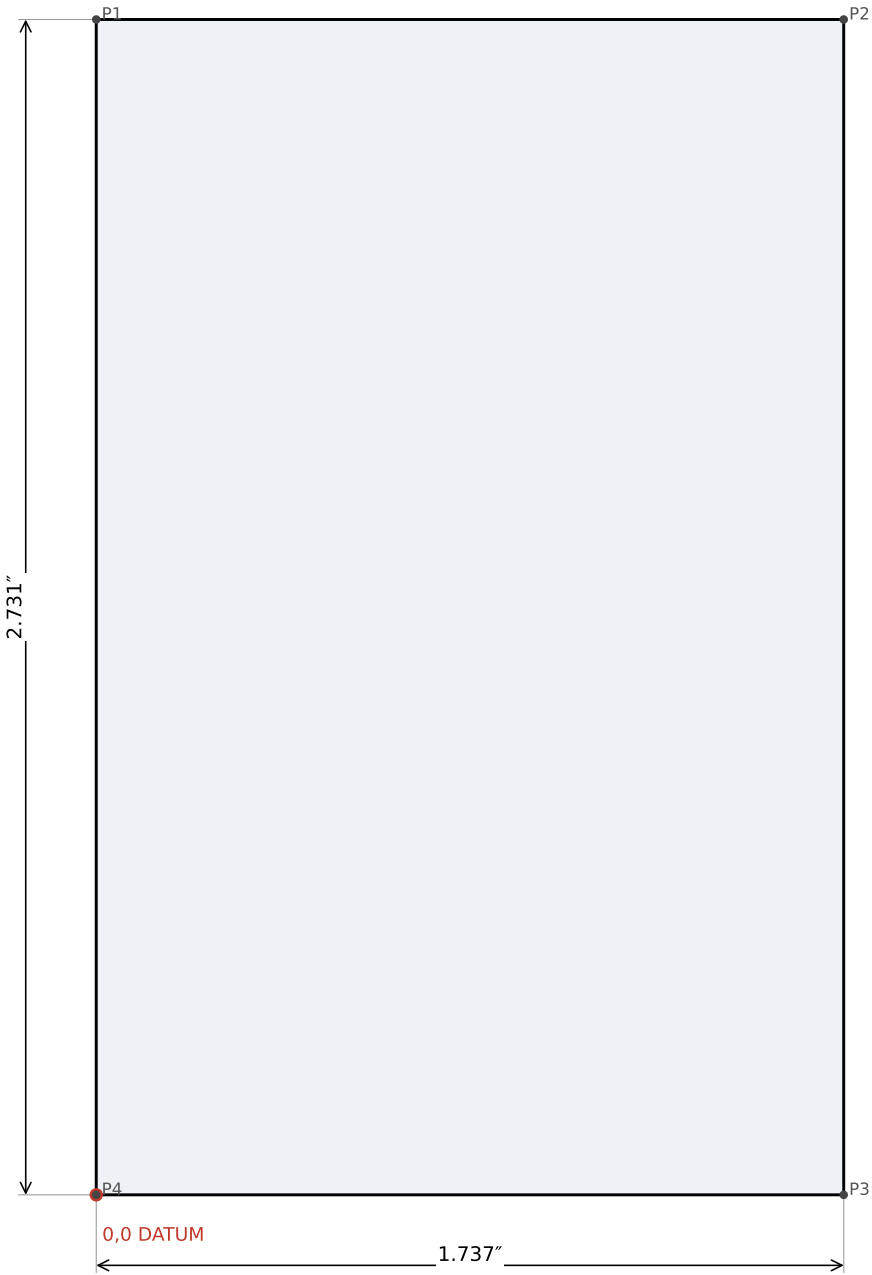
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 4 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate B-C	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 4 / 7

FR-10 (welded version) — blank 5 of 7

Plate E-F · blank 1.737" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	2.731
P2	1.737	2.731
P3	1.737	0.000
P4	0.000	0.000

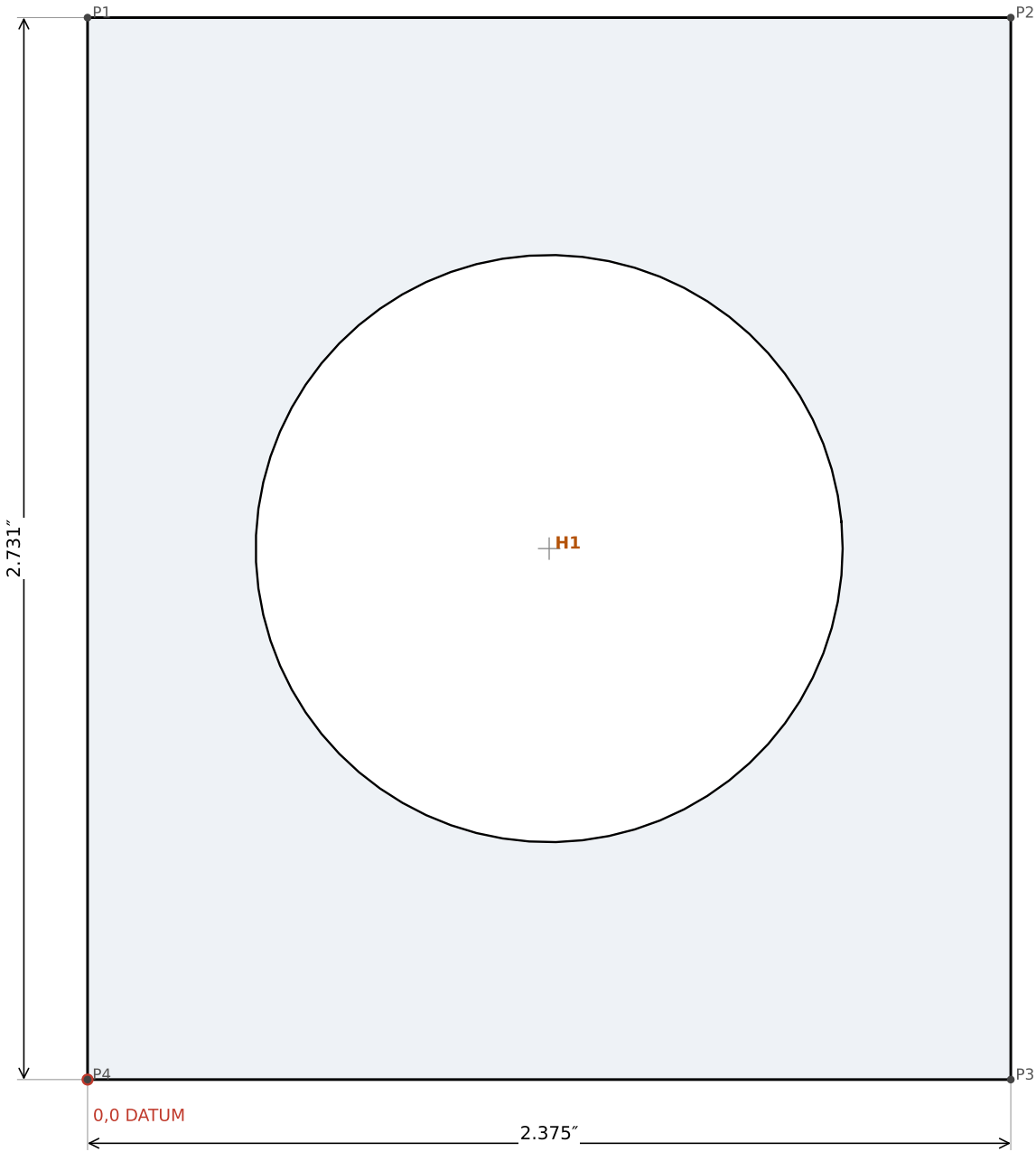
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 5 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate E-F	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 5 / 7

FR-10 (welded version) — blank 6 of 7

Shelf (hole Ø1.5 for the tube) · blank 2.375" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.510" X 1.187 Y 1.366

OUTLINE POINTS (X, Y from datum)

P1 0.000 2.731
P2 2.375 2.731
P3 2.375 0.000
P4 0.000 0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * + tube OD1.5 / ID1.181, 3.42", through A-B and the shelf, weld all round, cut off the excess

FR-10 (welded version) — blank 6 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Shelf (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 6 / 7

FR-10 (welded version) — blank 7 of 7

Inner plate · blank 5.748" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	2.731
P3	5.748	2.731
P4	5.748	0.000



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-10 (welded version) — blank 7 of 7	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Inner plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 7 / 7

FR-2 (bent version) — blank 1 of 1

FR-2 one piece (top + 2 U legs) · blank 22.078" × 3.250" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

H1 slot 0.709 × 0.472" @0° X 8.539 Y 1.500
H2 slot 0.750 × 0.625" @0° X 13.539 Y 1.375

DRILL AFTER WELDING (red dashed — do NOT laser cut)

D1 slot 0.709 × 0.472" X 4.007 Y 1.500
D2 slot 0.750 × 0.625" X 18.072 Y 1.375

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1 (1) 90° ↓ (2.999, 0.000) → (2.999, 3.250)
B2 (3) 90° ↓ (4.765, 0.000) → (4.765, 3.250)
B3 (5) 90° ↓ (7.781, 0.250) → (7.781, 3.250)
B4 (2) 90° ↓ (19.080, 0.000) → (19.080, 3.250)
B5 (4) 90° ↓ (17.314, 0.000) → (17.314, 3.250)
B6 (6) 90° ↓ (14.297, 0.250) → (14.297, 3.250)

OUTLINE POINTS (X, Y from datum)

P1 19.080 0.000
P2 18.697 0.000
P3 18.697 0.500
P4 17.697 0.500
P5 17.697 0.000
P6 17.314 0.000
P7 14.297 0.250
P8 7.781 0.250
P9 4.765 0.000
P10 4.382 0.000
P11 4.382 0.500
P12 3.382 0.500
P13 3.382 0.000
P14 2.999 0.000
P15 0.000 0.240
P16 0.000 3.250
P17 22.078 3.250
P18 22.078 0.240

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * one blank, 6 bends 90° the same way (R = t): inner walls first, then the bottoms, then the outer walls; top ovals laser cut; do NOT cut the leg-bottom ovals — drill after welding through the top ovals; ½" notch at the front edge of

FR-2 (bent version) — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FR-2 one piece (top + 2 U legs)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-2 (welded version) — blank 1 of 4

Top (2 slots) · blank 6.750" × 3.000" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)			
H1	slot	0.749 × 0.625"	@0° X 5.875 Y 1.125
H2	slot	0.709 × 0.472"	@0° X 0.875 Y 1.250
OUTLINE POINTS (X, Y from datum)			
P1	6.750	0.000	
P2	0.000	0.000	
P3	0.000	3.000	
P4	6.750	3.000	



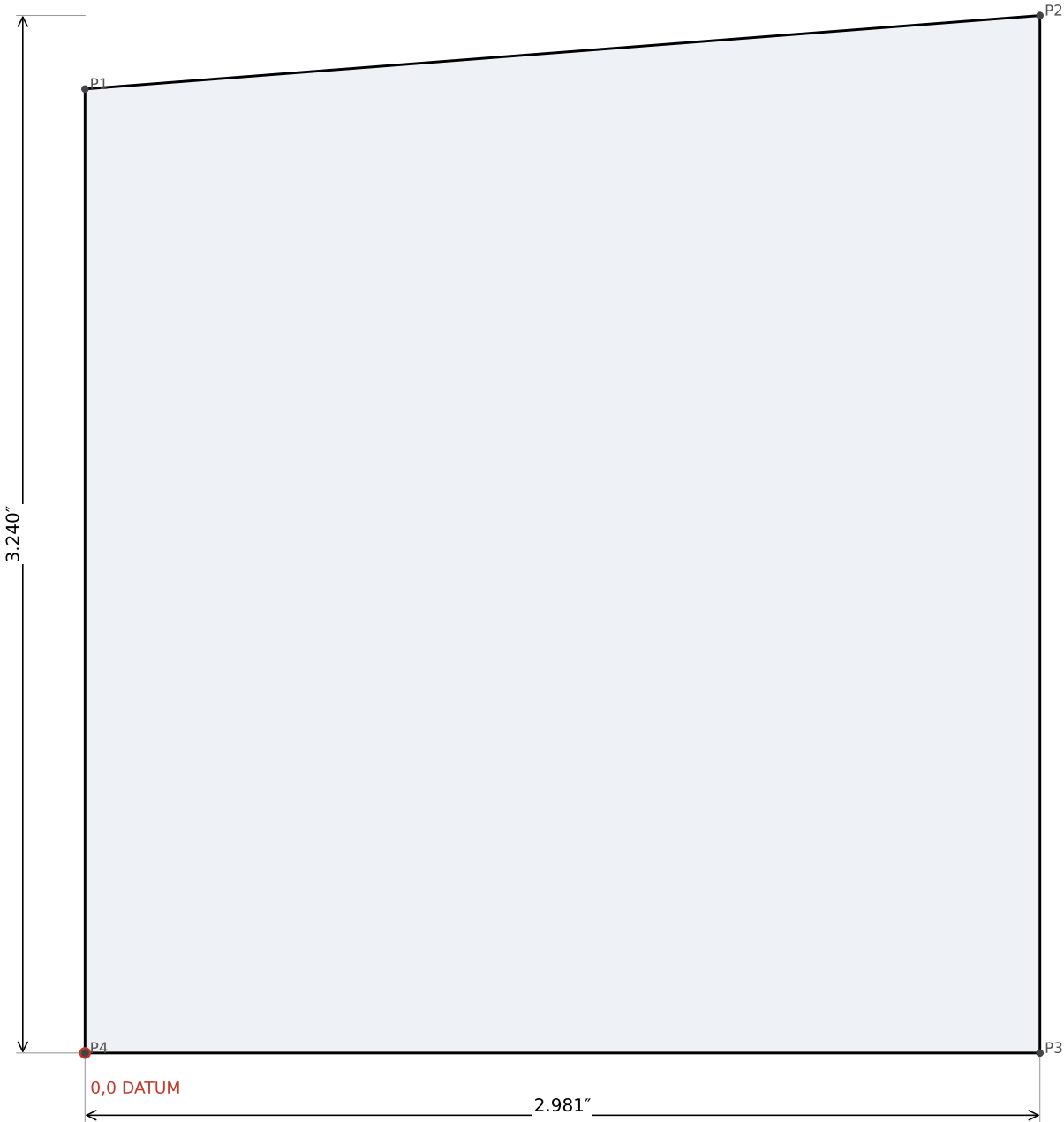
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 1 of 4	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top (2 slots)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FR-2 (welded version) — blank 2 of 4

Leg wall · blank 2.981" × 3.240" · HR steel sheet 10 ga (0.135") · 4 per side · FR + FL (mirror) = 8 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	3.010
P2	2.981	3.240
P3	2.981	0.000
P4	0.000	0.000

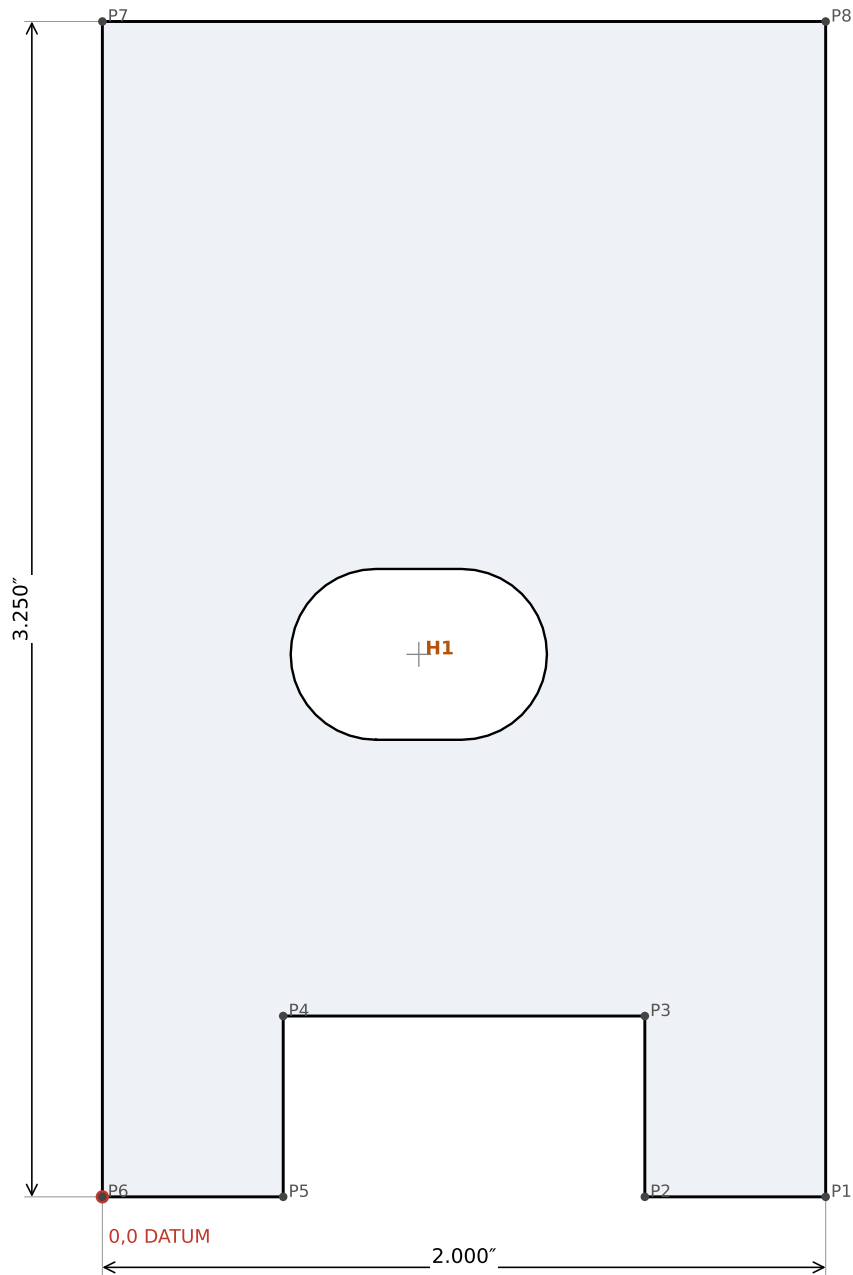
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 2 of 4	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg wall	QTY: 4 per side · FR + FL (mirror) = 8 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 4

FR-2 (welded version) — blank 3 of 4

Leg bottom, left (slot 0.709×0.472) · blank 2.000" × 3.250" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)		
H1	slot 0.709 × 0.472" @0°	X 0.875 Y 1.500
OUTLINE POINTS (X, Y from datum)		
P1	2.000	0.000
P2	1.500	0.000
P3	1.500	0.500
P4	0.500	0.500
P5	0.500	0.000
P6	0.000	0.000
P7	0.000	3.250
P8	2.000	3.250

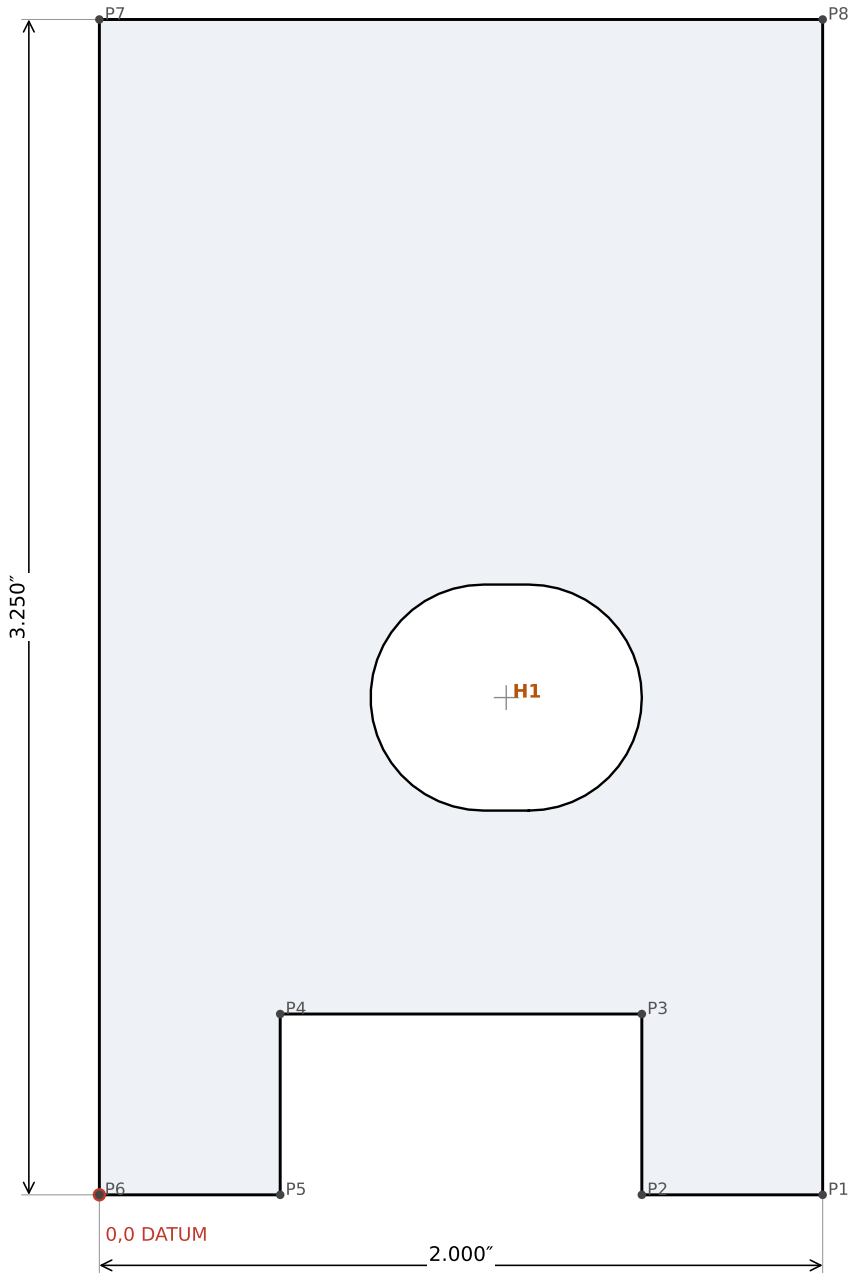
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 3 of 4	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg bottom, left (slot 0.709×0.472)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FR-2 (welded version) — blank 4 of 4

Leg bottom, right (slot 3/4×5/8) · blank 2.000″ × 3.250″ · HR steel sheet 10 ga (0.135″) · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)		
H1	slot 0.749 × 0.625″ @0°	X 1.125 Y 1.375
OUTLINE POINTS (X, Y from datum)		
P1	2.000	0.000
P2	1.500	0.000
P3	1.500	0.500
P4	0.500	0.500
P5	0.500	0.000
P6	0.000	0.000
P7	0.000	3.250
P8	2.000	3.250

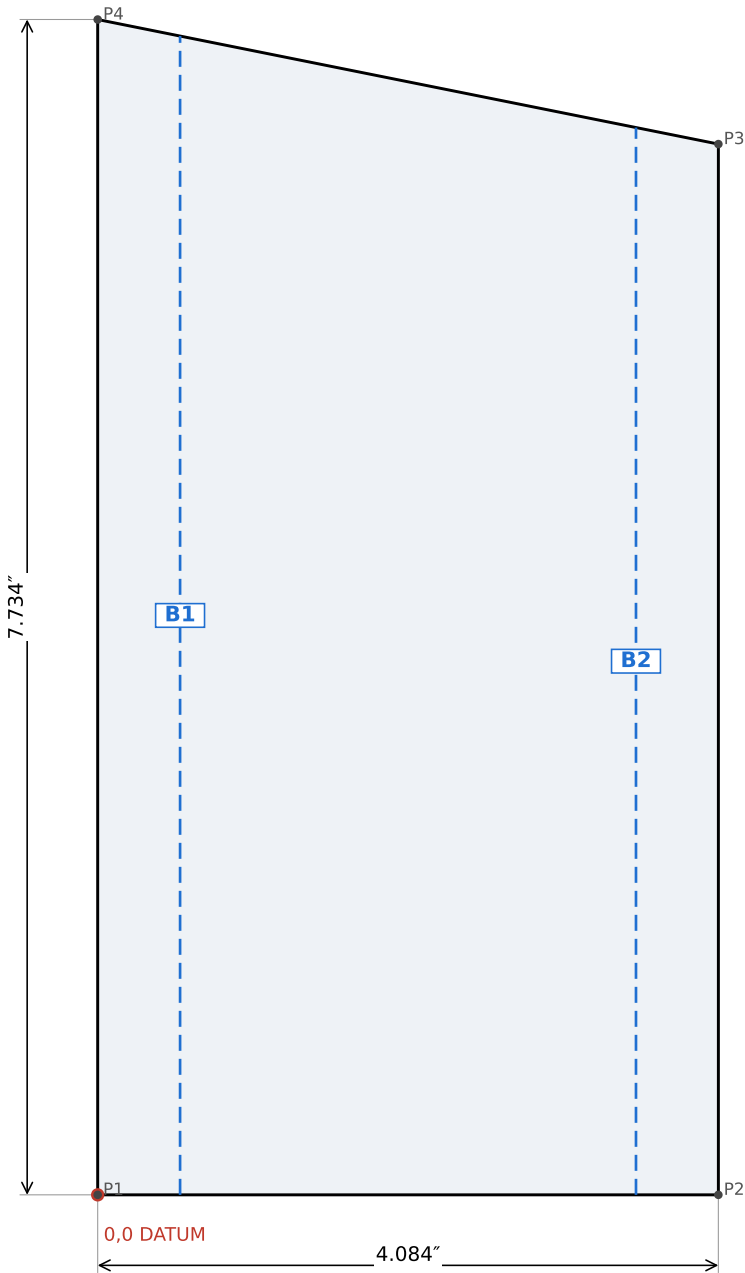
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135″) (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-2 (welded version) — blank 4 of 4	HR steel sheet 10 ga (0.135″)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Leg bottom, right (slot 3/4×5/8)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 4 / 4

FR-3 — blank 1 of 1

Channel — flat pattern (2 bends 90°) · blank 4.084" × 7.734" · HR steel sheet 11 ga (0.120") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 11 ga (0.120")

B1 (1) 90° (0.542, 0.000) → (0.542, 7.625)
B2 (2) 90° (3.542, 0.000) → (3.542, 7.023)

OUTLINE POINTS (X, Y from datum)

P1 0.000 0.000
P2 4.084 0.000
P3 4.084 6.914
P4 0.000 7.734

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 11 ga (0.120") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend 90° on dashed lines, 3/4 flanges inward

FR-3 — blank 1 of 1	HR steel sheet 11 ga (0.120")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel — flat pattern (2 bends 90°)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-4 (bent version) — blank 1 of 3

Wrap strip (ear, top, front, bottom, slant) · blank 18.746" × 2.716" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

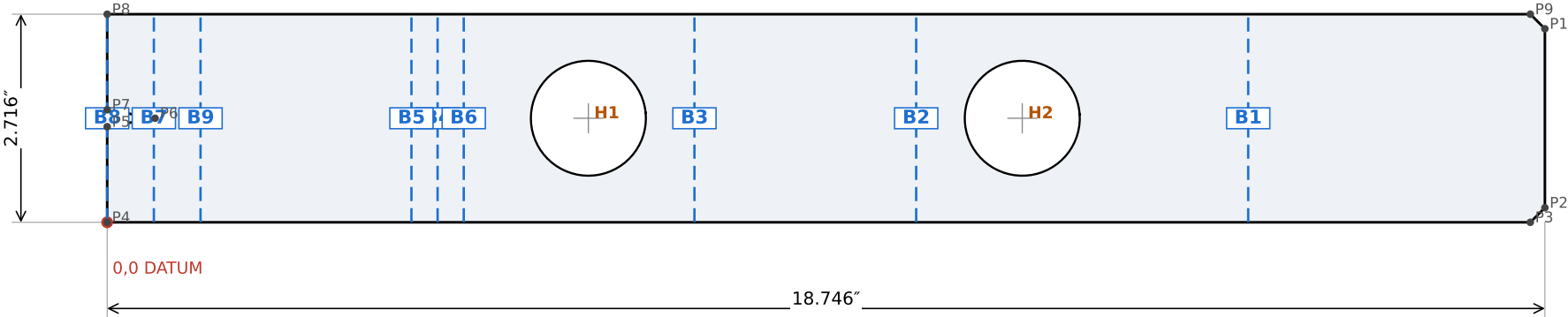
H1	Ø1.500"	X 6.274	Y 1.358
H2	Ø1.500"	X 11.932	Y 1.358

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1	(5) 60°	(14.878, 0.000) → (14.878, 2.716)
B2	(4) 90°	(10.549, 0.000) → (10.549, 2.716)
B3	(3) 90°	(7.657, 0.000) → (7.657, 2.716)
B4	(2) 90° ↓ R0.38	(4.308, 0.000) → (4.308, 2.716)
B5	radius-bend edge	(3.967, 0.000) → (3.967, 2.716)
B6	radius-bend edge	(4.649, 0.000) → (4.649, 2.716)
B7	(1) 126.9° R0.49	(0.609, 0.000) → (0.609, 2.716)
B8	radius-bend edge	(0.000, 0.000) → (0.000, 2.716)
B9	radius-bend edge	(1.217, 0.000) → (1.217, 2.716)
radius-bend edge = start/end of a large-radius bend (bump bend between them)		

OUTLINE POINTS (X, Y from datum)

P1	18.746	2.526
P2	18.746	0.190
P3	18.556	0.000
P4	0.000	0.000
P5	0.000	1.248
P6	0.625	1.358
P7	0.000	1.468
P8	0.000	2.716
P9	18.556	2.716



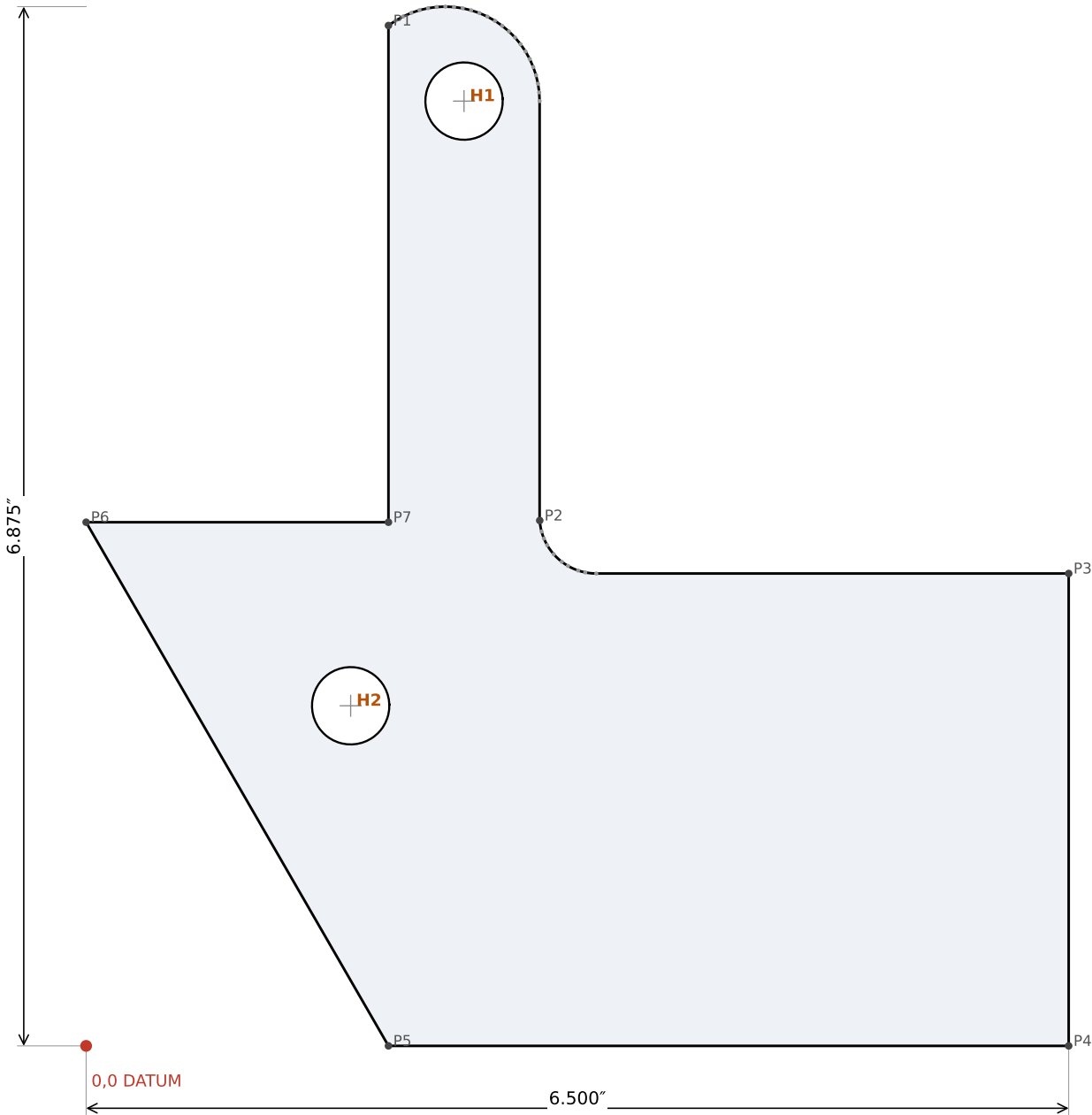
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * arc C-D bent over R 0.48; R ⅜" at E; no corner reliefs

FR-4 (bent version) — blank 1 of 3	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Wrap strip (ear, top, front, bottom, slant)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 4

FR-4 (bent version) — blank 2 of 3

Side plate FR-4 (holes 0.512) · blank 6.500" × 6.875" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.512"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.001	3.475
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.048	6.782
2.098	6.810
2.151	6.833
2.206	6.852
2.262	6.865
2.319	6.872
2.377	6.875
2.434	6.872
2.491	6.864
2.547	6.851
2.602	6.832
2.655	6.809
2.705	6.781
2.752	6.748
2.797	6.711
2.837	6.671
2.874	6.626
2.907	6.579
2.935	6.528
2.958	6.476
2.976	6.421
2.989	6.365
2.997	6.308
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.013	3.403
3.029	3.357
3.050	3.312
3.093	3.253
3.147	3.203
3.188	3.175
3.255	3.145
3.302	3.132
3.375	3.125

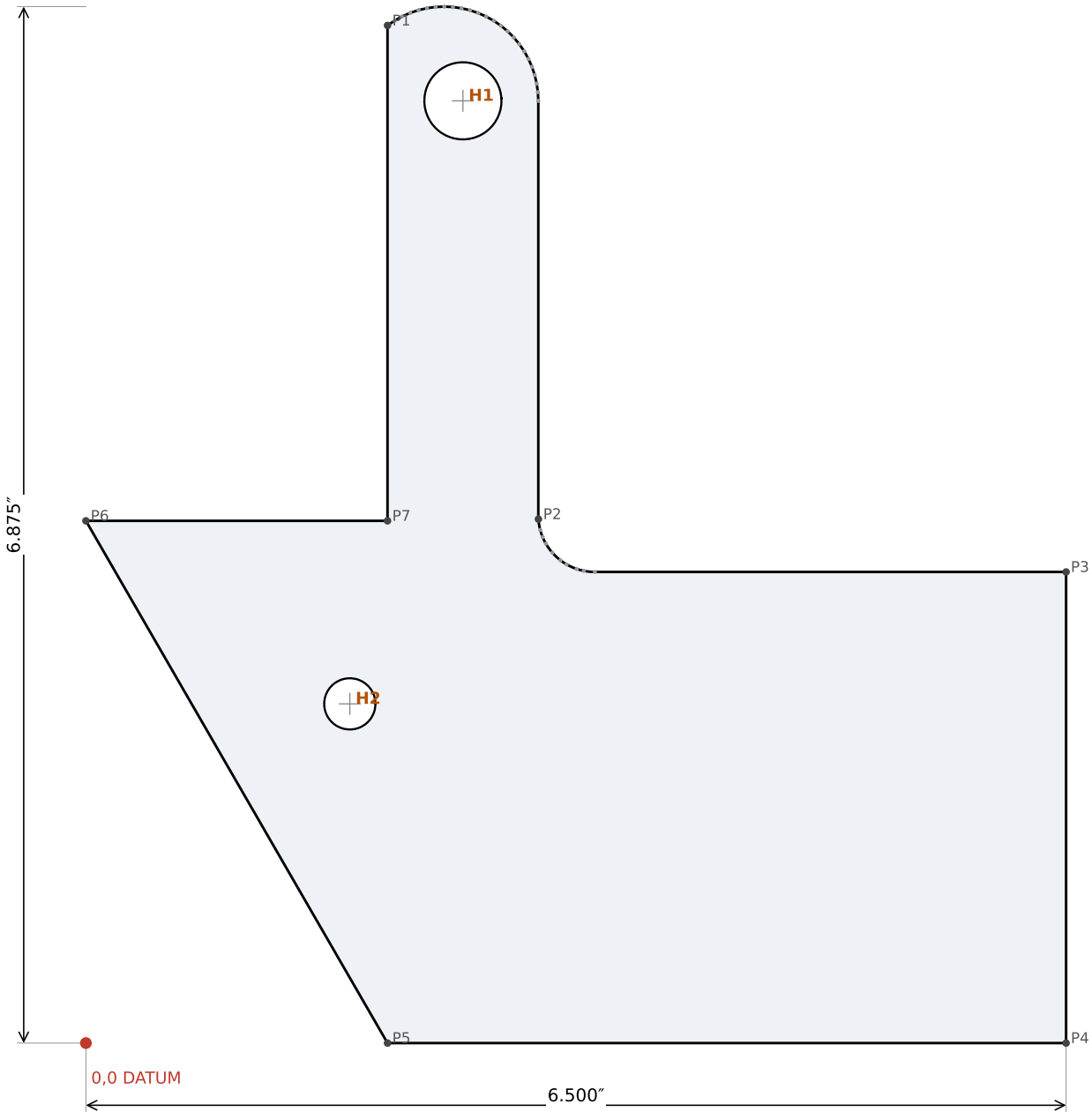
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-4 (bent version) — blank 2 of 3	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate FR-4 (holes 0.512)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 4

FR-4 (bent version) — blank 3 of 3

Side plate FL-4 (holes 0.512 and 0.34) · blank 6.500” × 6.875” · HR steel sheet 10 ga (0.135”) · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.340"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.001	3.475
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.048	6.782
2.098	6.810
2.151	6.833
2.206	6.852
2.262	6.865
2.319	6.872
2.377	6.875
2.434	6.872
2.491	6.864
2.547	6.851
2.602	6.832
2.655	6.809
2.705	6.781
2.752	6.748
2.797	6.711
2.837	6.671
2.874	6.626
2.907	6.579
2.935	6.528
2.958	6.476
2.976	6.421
2.989	6.365
2.997	6.308
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.013	3.403
3.029	3.357
3.050	3.312
3.093	3.253
3.147	3.203
3.188	3.175
3.255	3.145
3.302	3.132
3.375	3.125

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135”) (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * flat, no bends

FR-4 (bent version) — blank 3 of 3	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate FL-4 (holes 0.512 and 0.34)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 4

FR-4 (bent version) — cut list (tubes)

• Tube OD 1.5 / ID 1.181 — cut 3.689” (through bottom and top)

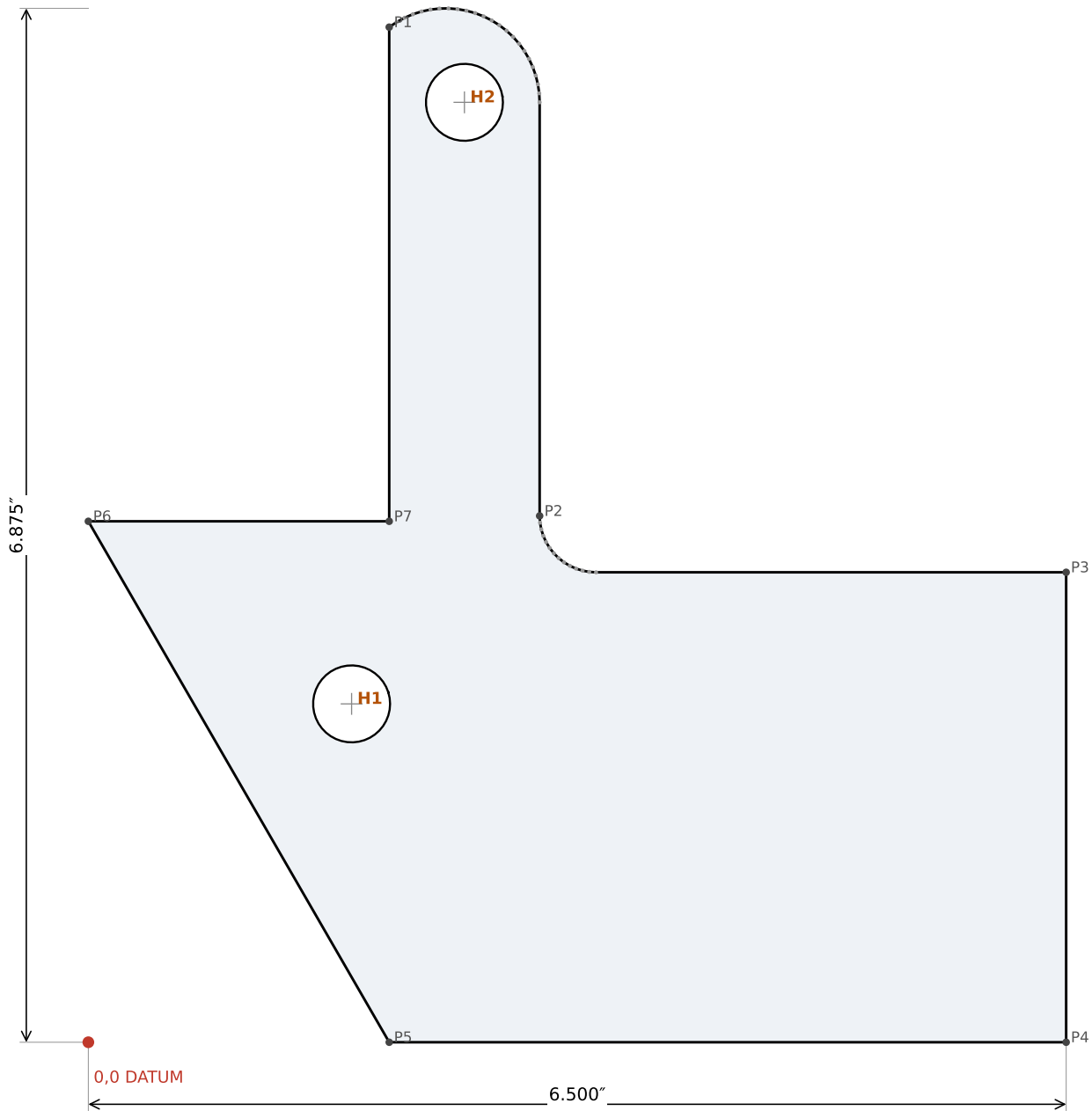
Stock: DOM round tube · cut length 3.689” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-4 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube OD 1.5 / ID 1.181 — cut 3.689” (through bottom and top)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 4 / 4

FR-4 (welded version) — blank 1 of 8

Side plate, right (hole Ø0.512) · blank 6.500" × 6.875" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 1.750	Y 2.250
H2	Ø0.512"	X 2.500	Y 6.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.000	3.500
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.050	6.784
2.103	6.812
2.158	6.836
2.215	6.854
2.274	6.867
2.334	6.874
2.394	6.875
2.454	6.870
2.513	6.859
2.571	6.843
2.627	6.822
2.681	6.795
2.732	6.763
2.780	6.726
2.824	6.685
2.863	6.640
2.899	6.591
2.929	6.539
2.954	6.485
2.974	6.428
2.988	6.370
2.997	6.310
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.003	3.455
3.011	3.410
3.024	3.367
3.043	3.326
3.066	3.287
3.094	3.251
3.126	3.219
3.162	3.191
3.201	3.168
3.242	3.149
3.285	3.136
3.330	3.128
3.375	3.125

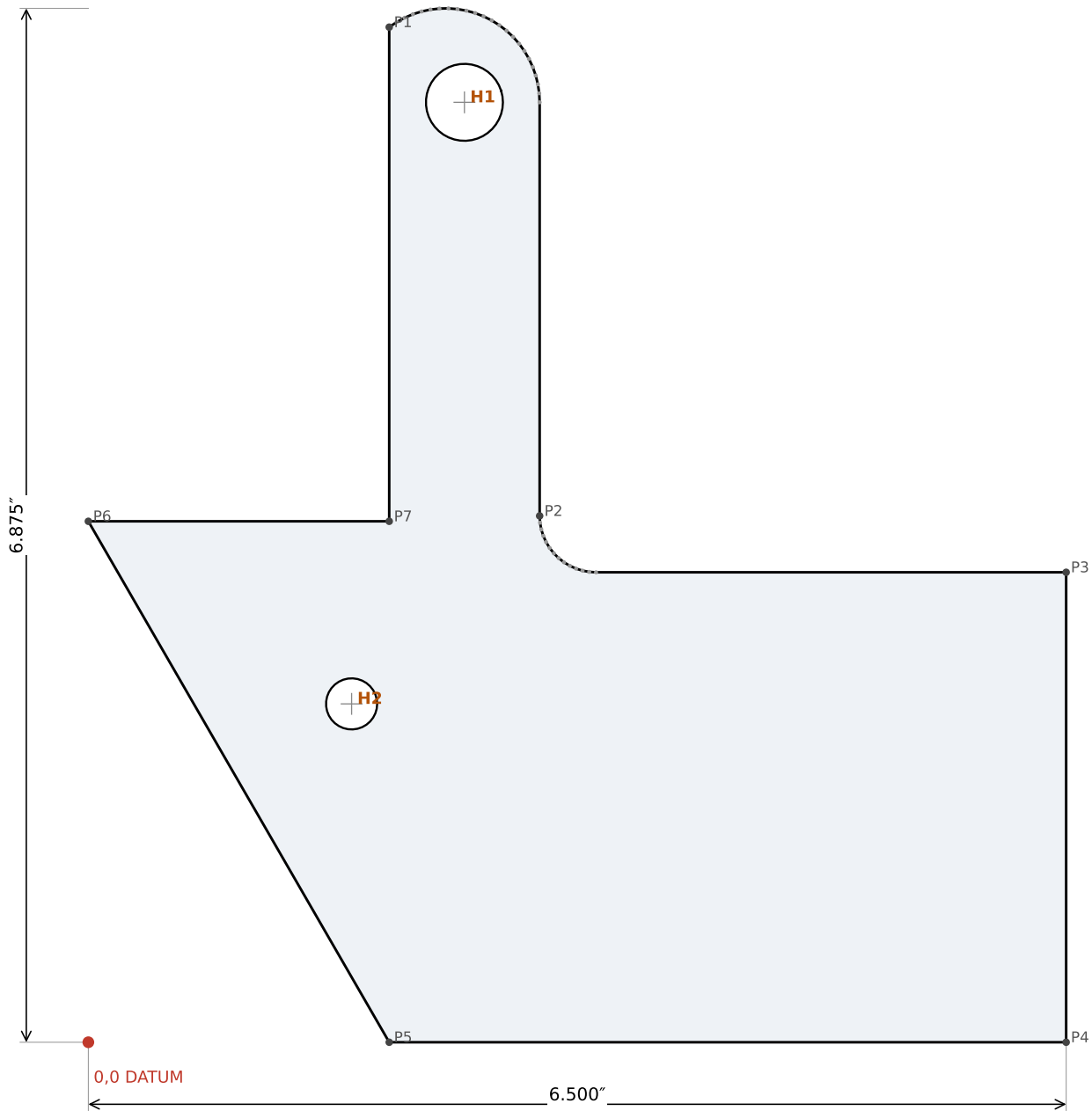
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 1 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate, right (hole Ø0.512)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 8

FR-4 (welded version) — blank 2 of 8

Side plate, left (hole Ø0.34) · blank 6.500" × 6.875" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1	Ø0.512"	X 2.500	Y 6.250
H2	Ø0.340"	X 1.750	Y 2.250

OUTLINE POINTS (X, Y from datum)

P1	2.000	6.750
P2	3.000	3.500
P3	6.500	3.125
P4	6.500	0.000
P5	2.000	0.000
P6	0.000	3.464
P7	2.000	3.464

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.050	6.784
2.103	6.812
2.158	6.836
2.215	6.854
2.274	6.867
2.334	6.874
2.394	6.875
2.454	6.870
2.513	6.859
2.571	6.843
2.627	6.822
2.681	6.795
2.732	6.763
2.780	6.726
2.824	6.685
2.863	6.640
2.899	6.591
2.929	6.539
2.954	6.485
2.974	6.428
2.988	6.370
2.997	6.310
3.000	6.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.003	3.455
3.011	3.410
3.024	3.367
3.043	3.326
3.066	3.287
3.094	3.251
3.126	3.219
3.162	3.191
3.201	3.168
3.242	3.149
3.285	3.136
3.330	3.128
3.375	3.125

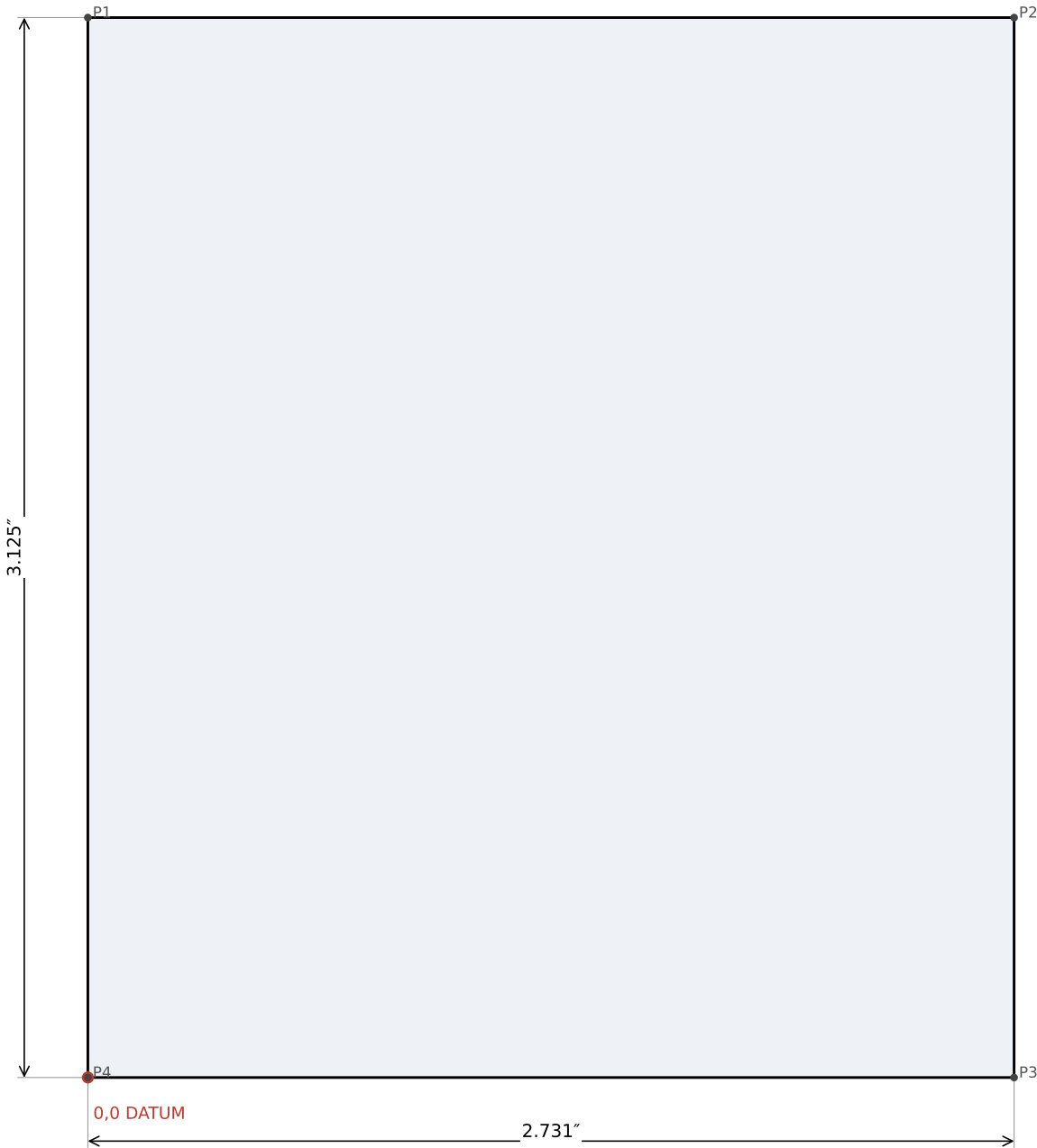
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
3. Flat pattern already includes bend allowance (K = 0.44, R = t).
4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
5. Deburr all edges. Curved edges: smooth arcs per DXF.
6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 2 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate, left (hole Ø0.34)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 8

FR-4 (welded version) — blank 3 of 8

Front plate · blank 2.731" × 3.125" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	3.125
P2	2.731	3.125
P3	2.731	0.000
P4	0.000	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 3 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Front plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 8

FR-4 (welded version) — blank 4 of 8

Slanted plate · blank 4.000" × 2.731" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	2.731
P3	4.000	2.731
P4	4.000	0.000

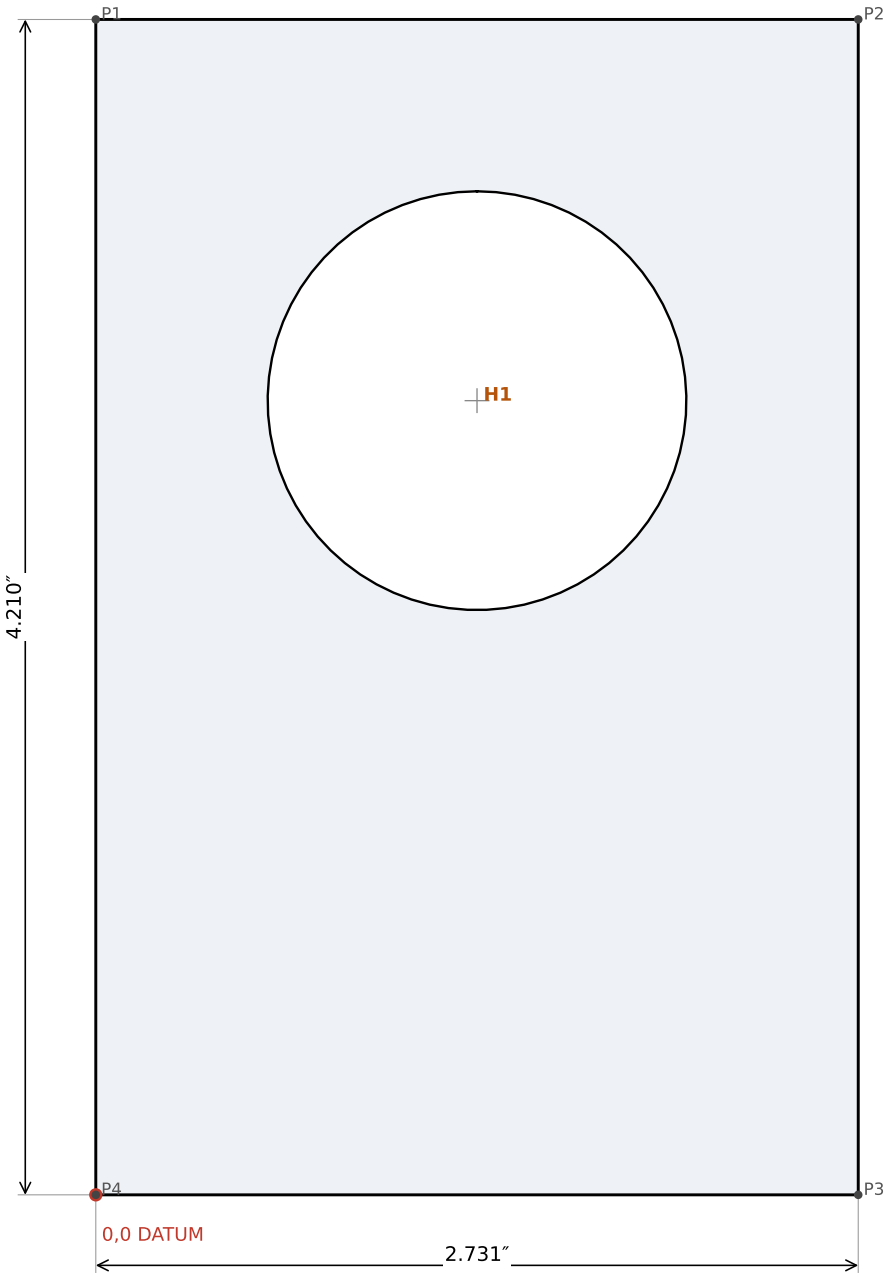
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 4 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Slanted plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 4 / 8

FR-4 (welded version) — blank 5 of 8

Bottom plate (hole Ø1.5 for the tube) · blank 2.731” × 4.210” · HR steel sheet 10 ga (0.135”) · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.500” X 1.366 Y 2.845

OUTLINE POINTS (X, Y from datum)

P1 0.000 4.210
P2 2.731 4.210
P3 2.731 0.000
P4 0.000 0.000

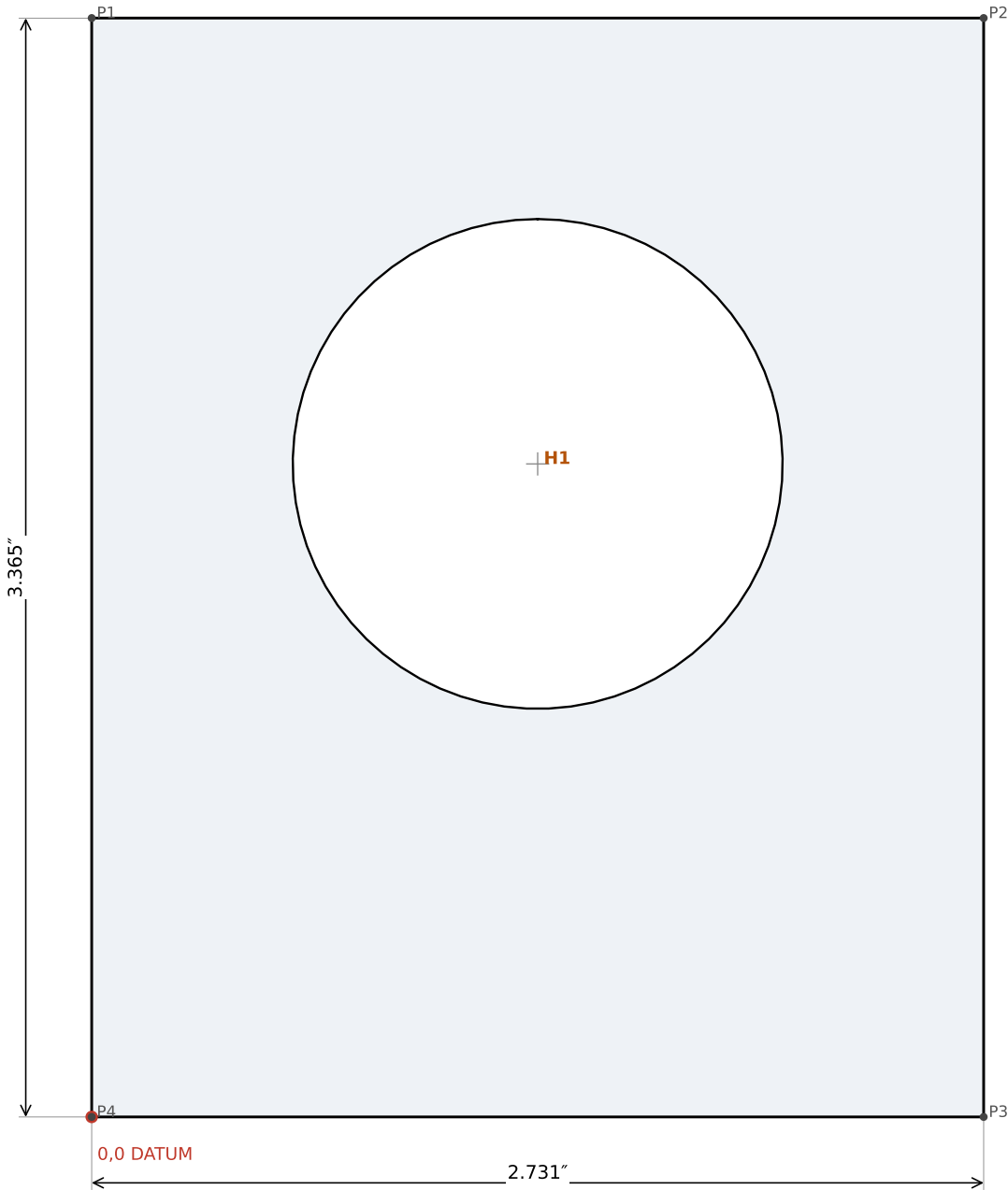
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135”) (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * + tube OD1.5 / ID1.181 through (bottom + top), weld all round

FR-4 (welded version) — blank 5 of 8	HR steel sheet 10 ga (0.135”)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Bottom plate (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 5 / 8

FR-4 (welded version) — blank 6 of 8

Top plate (hole Ø1.5 for the tube) · blank 2.731” × 3.365” · HR steel sheet 10 ga (0.135”) · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø1.500" X 1.365 Y 2.000

OUTLINE POINTS (X, Y from datum)

P1 0.000 3.365
P2 2.731 3.365
P3 2.731 0.000
P4 0.000 0.000

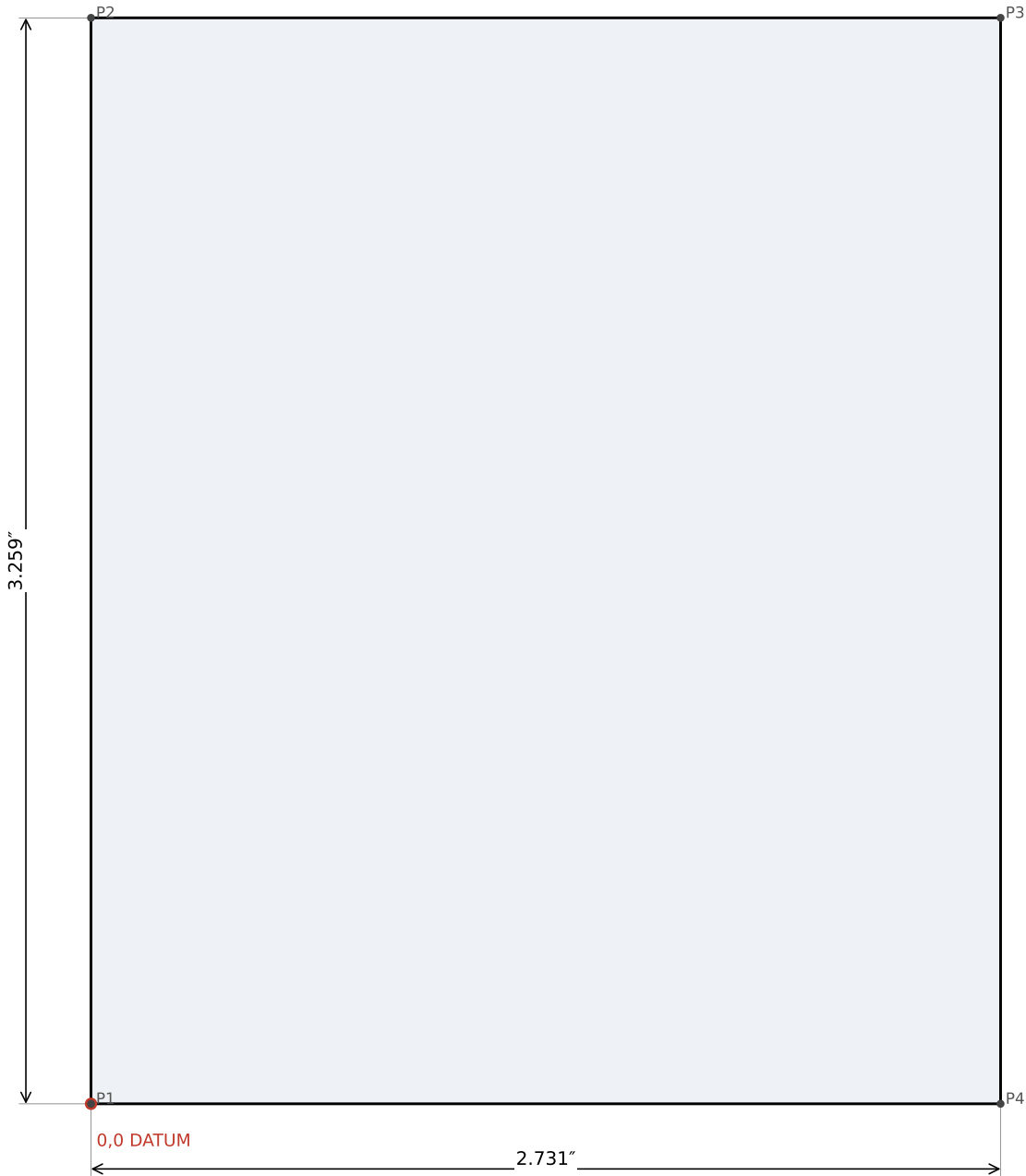
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135”) (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * the tube passes through

FR-4 (welded version) — blank 6 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top plate (hole Ø1.5 for the tube)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 6 / 8

FR-4 (welded version) — blank 7 of 8

Ear plate · blank 2.731" × 3.259" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.260
P3	2.731	3.260
P4	2.731	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-4 (welded version) — blank 7 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Ear plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 7 / 8

FR-4 (welded version) — blank 8 of 8

Ear cap (bend R0.625) · blank 2.705" × 1.200" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle

HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	2.705	0.000
P2	2.705	1.200
P3	0.000	1.200
P4	0.000	0.000



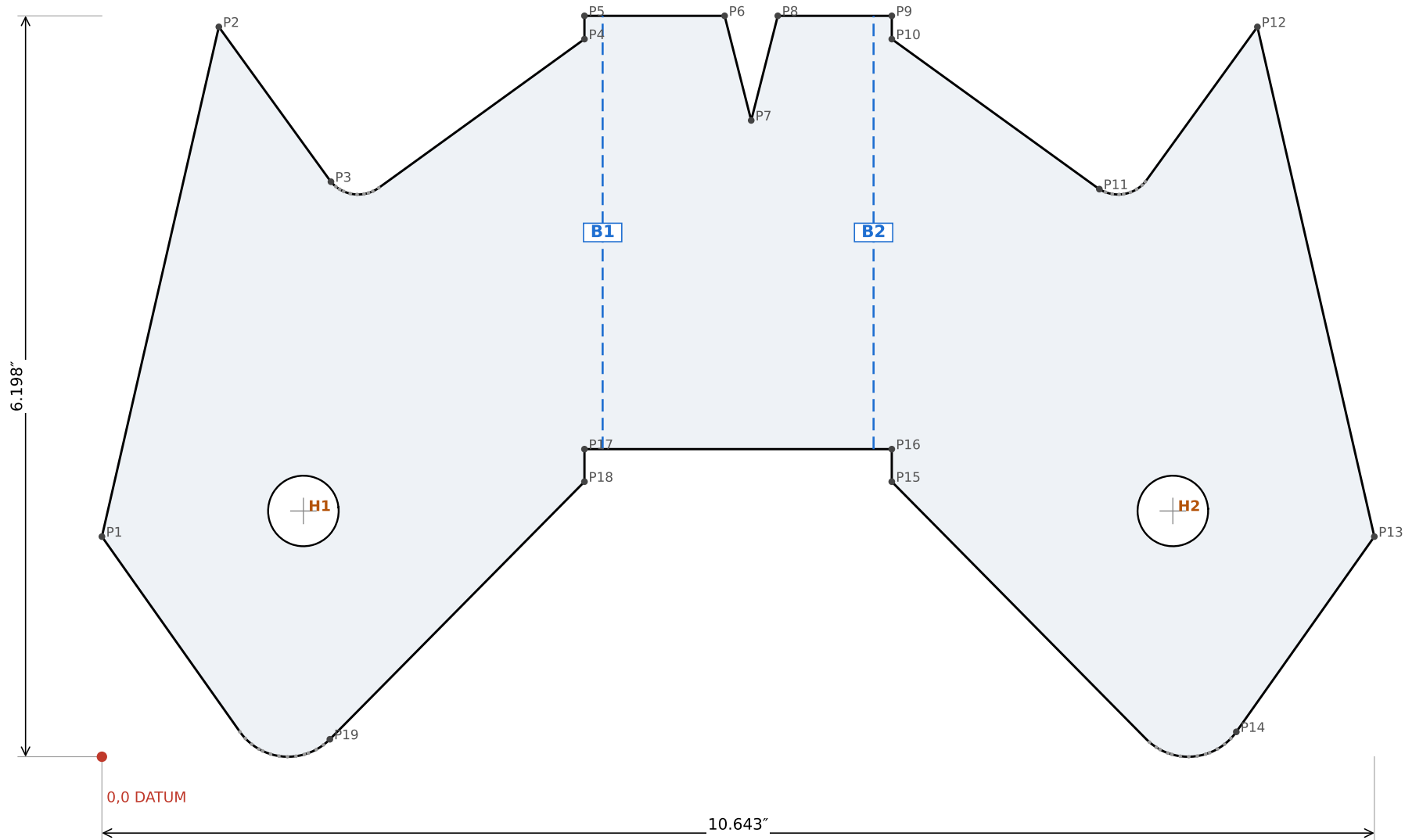
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * bend to arc R 0.625; V-notch 5/8

FR-4 (welded version) — blank 8 of 8	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Ear cap (bend R0.625)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 8 / 8

FR-5 (bent version) — blank 1 of 1

U-body (2 sides + plate A-C) · blank 10.643” × 6.198” · HR steel sheet 10 ga (0.135”) · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135”) (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t

HOLES (centre X, Y from datum)

H1 Ø0.591” X 1.685 Y 2.056
H2 Ø0.591” X 8.957 Y 2.056

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135”)

B1 (1) 90° (4.188, 2.573) → (4.188, 6.198)
B2 (2) 90° (6.454, 2.573) → (6.454, 6.198)

OUTLINE POINTS (X, Y from datum)

P1 0.000 1.842
P2 0.978 6.106
P3 1.916 4.810
P4 4.036 6.004
P5 4.036 6.198
P6 5.208 6.198
P7 5.431 5.323
P8 5.653 6.198
P9 6.607 6.198
P10 6.607 6.004
P11 8.343 4.749
P12 9.665 6.106
P13 10.643 1.842
P14 9.488 0.208
P15 6.607 2.301
P16 6.607 2.573
P17 4.036 2.573
P18 4.036 2.301
P19 1.907 0.146

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

1.957 4.770
1.988 4.747
2.022 4.729
2.077 4.711
2.135 4.704
2.192 4.708
2.230 4.717
2.266 4.730
2.316 4.760

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

8.395 4.723
8.450 4.708
8.508 4.704
8.546 4.707
8.602 4.722
8.654 4.747
8.700 4.782
8.727 4.810

FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y

9.443 0.153
9.390 0.105
9.330 0.065
9.266 0.034
9.232 0.022
9.162 0.006
9.091 0.000
9.019 0.004
8.949 0.019
8.915 0.031
8.850 0.060
8.819 0.079
8.762 0.122

FREE-FORM CURVE 4 (grey dots, follow DXF) — points X, Y

1.853 0.099
1.793 0.060

FR-5 (bent version) — blank 1 of 1	HR steel sheet 10 ga (0.135”)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
U-body (2 sides + plate A-C)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-5 (bent version) — blank 1 of 1 — tables (continued)

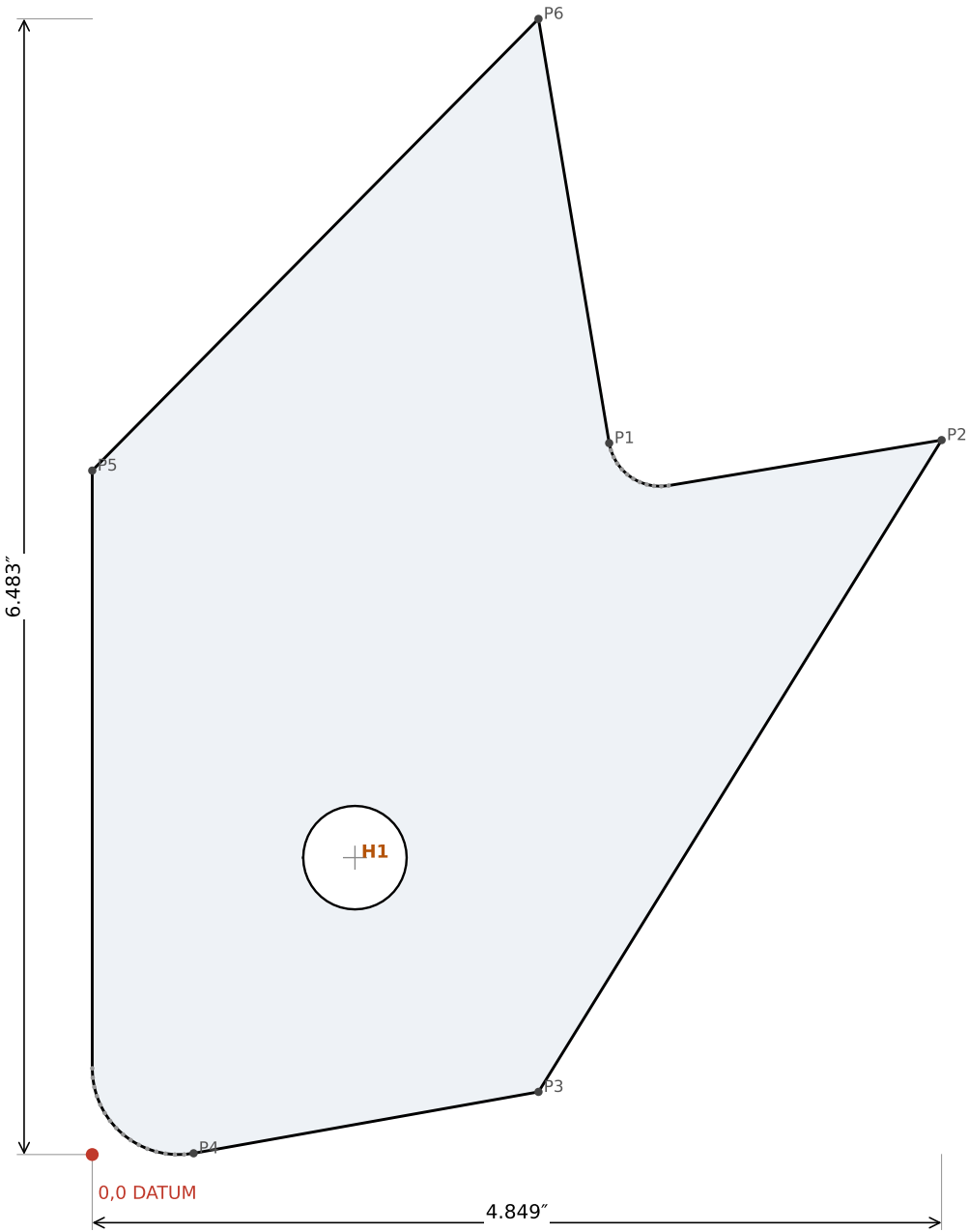
1.728 0.031
1.694 0.019
1.623 0.004
1.552 0.000
1.480 0.006
1.411 0.022

1.344 0.048
1.312 0.065
1.253 0.105
1.200 0.153
1.154 0.208

FR-5 (bent version) — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
U-body (2 sides + plate A-C)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-5 (welded version) — blank 1 of 2

Side plate · blank 4.849" × 6.483" · HR steel sheet 10 ga (0.135") · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591" X 1.500 Y 1.695

OUTLINE POINTS (X, Y from datum)

P1 2.951 4.061
P2 4.849 4.078
P3 2.548 0.358
P4 0.579 0.007
P5 0.000 3.904
P6 2.548 6.483

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

2.961 4.020
2.977 3.981
2.998 3.945
3.024 3.911
3.054 3.882
3.089 3.858
3.126 3.838
3.166 3.825
3.207 3.817
3.249 3.815
3.291 3.819

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

0.525 0.001
0.472 0.000
0.418 0.005
0.366 0.016
0.315 0.033
0.266 0.055
0.219 0.082
0.176 0.115
0.137 0.151
0.102 0.192
0.072 0.236
0.046 0.284
0.026 0.334
0.012 0.385
0.003 0.438
0.000 0.492

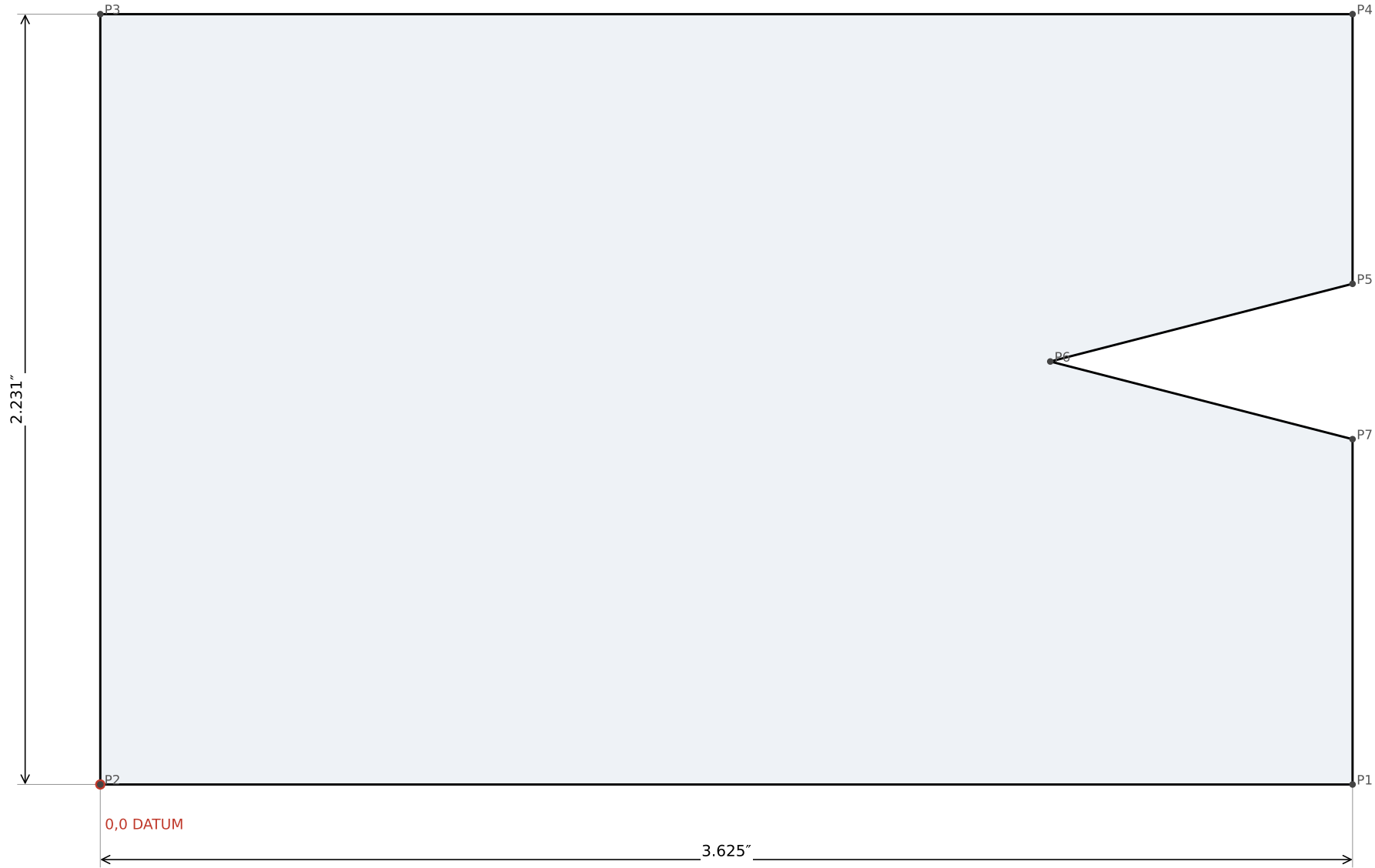
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-5 (welded version) — blank 1 of 2	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 2

FR-5 (welded version) — blank 2 of 2

Plate on edge A-C · blank 3.625" × 2.231" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	3.625	0.000
P2	0.000	0.000
P3	0.000	2.231
P4	3.625	2.231
P5	3.625	1.450
P6	2.750	1.225
P7	3.625	1.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-5 (welded version) — blank 2 of 2	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Plate on edge A-C	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 2

FR-6 — cut list (tubes)

• **Tube 3×3 — cut 14 ¼”**

Stock: Square tube 3” × 3”, wall 3/16” · cut length 14.250” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

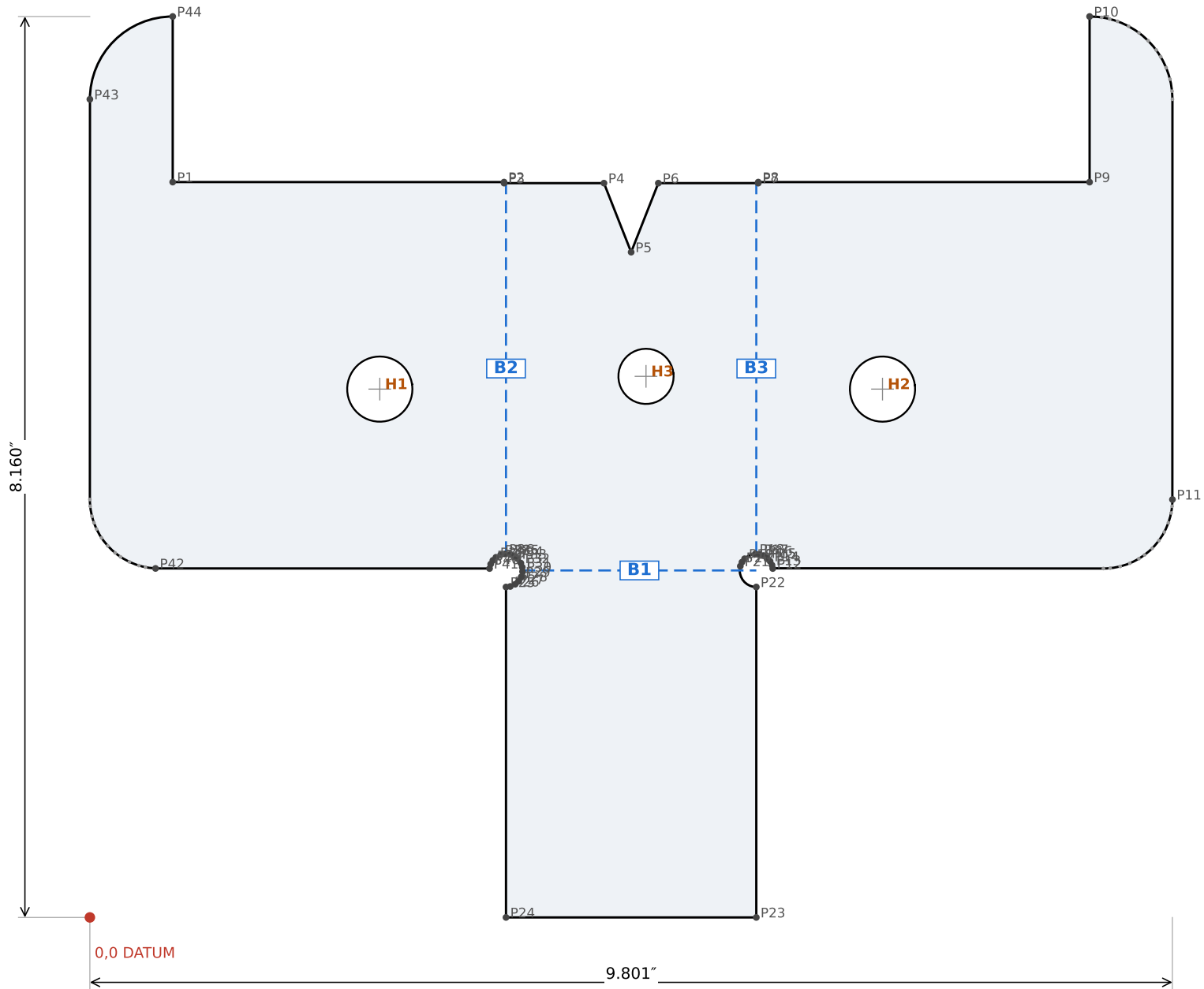
wall 3/16”

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-6 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube 3×3 — cut 14 ¼”	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1

FR-7 (bent version) — blank 1 of 1

FR-7 one piece (back + bottom + 2 sides) · blank 9.801" × 8.160" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).
- * one blank, 3 bends 90° the same way (R = t): bottom first, then the sides; Ø0.3 round reliefs in the two corners; after bending weld the side-to-bottom seams; V notch and Ø0.5 hole in the back, Ø0.591 holes in the sides

HOLES (centre X, Y from datum)

H1	Ø0.591"	X 2.625	Y 4.785
H2	Ø0.591"	X 7.176	Y 4.785
H3	Ø0.500"	X 5.035	Y 4.901

BENDS (bend centre line, from datum) IR = t = 10 ga (0.135")

B1	(1) 90°	(3.918, 3.143) → (6.034, 3.143)
B2	(2) 90°	(3.768, 3.293) → (3.768, 6.651)
B3	(3) 90°	(6.034, 3.293) → (6.034, 6.651)

CURVED EDGES (radius, centre X, Y)

R1	R 0.150"	centre 6.034, 3.143
R2	R 0.150"	centre 6.034, 3.143
R3	R 0.750"	centre 0.750, 7.410

OUTLINE POINTS (X, Y from datum)

P1	0.750	6.660
P2	3.750	6.660
P3	3.750	6.651
P4	4.655	6.651
P5	4.901	6.026
P6	5.147	6.651
P7	6.051	6.651
P8	6.051	6.660
P9	9.051	6.660
P10	9.052	8.160
P11	9.802	3.785
P12	6.183	3.160
P13	6.176	3.191
P14	6.159	3.226
P15	6.133	3.255
P16	6.100	3.277
P17	6.063	3.290
P18	6.024	3.292
P19	5.928	3.249
P20	5.904	3.218
P21	5.889	3.181
P22	6.034	2.993
P23	6.034	0.000
P24	3.768	0.000
P25	3.768	2.993
P26	3.806	2.998
P27	3.851	3.018
P28	3.880	3.044
P29	3.906	3.085
P30	3.917	3.133
P31	3.915	3.172
P32	3.902	3.209
P33	3.874	3.249
P34	3.843	3.272
P35	3.806	3.288
P36	3.758	3.292
P37	3.719	3.285
P38	3.676	3.262
P39	3.649	3.234
P40	3.629	3.200
P41	3.619	3.160
P42	0.594	3.161
P43	0.000	7.410
P44	0.750	8.160

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

9.162	8.152
9.234	8.138
9.304	8.116
9.405	8.072

FR-7 (bent version) — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FR-7 one piece (back + bottom + 2 sides)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

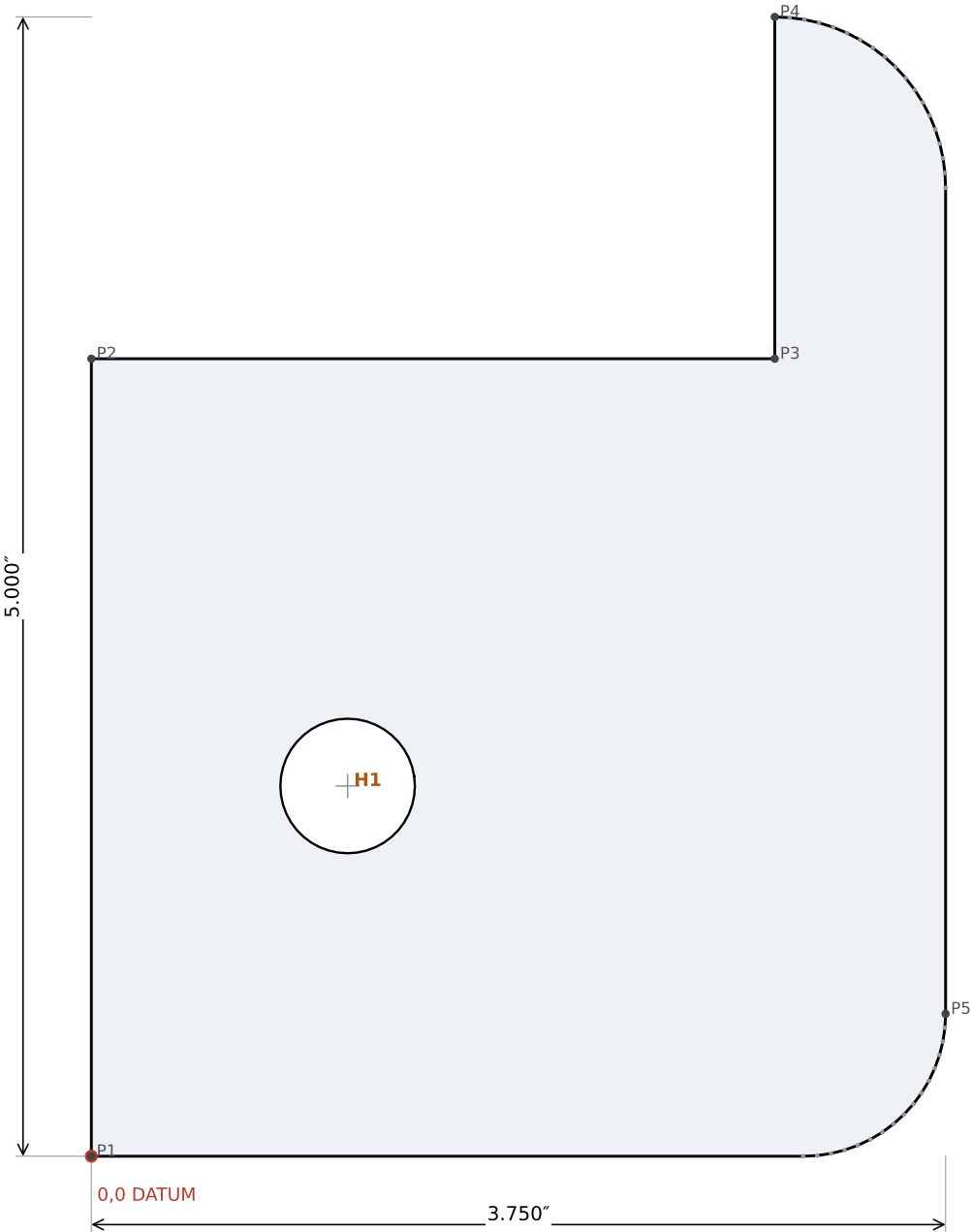
FR-7 (bent version) — blank 1 of 1 — tables (continued)

9.498	8.013	9.416	3.208
9.582	7.941	9.328	3.179
9.654	7.857	9.238	3.163
9.695	7.796	9.177	3.160
9.744	7.697	FREE-FORM CURVE 3 (grey dots, follow DXF) — points X, Y	
9.779	7.593	0.503	3.172
9.798	7.484	0.444	3.187
9.802	7.410	0.358	3.220
FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y		0.278	3.266
9.799	3.724	0.229	3.302
9.790	3.663	0.162	3.365
9.765	3.575	0.105	3.438
9.728	3.491	0.060	3.518
9.678	3.413	0.036	3.575
9.618	3.343	0.019	3.633
9.549	3.283	0.003	3.724
9.498	3.249	0.000	3.785

FR-7 (bent version) — blank 1 of 1	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
FR-7 one piece (back + bottom + 2 sides)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-7 (welded version) — blank 1 of 3

Side plate with tab · blank 3.750″ × 5.000″ · HR steel sheet 10 ga (0.135″) · 2 per side · FR + FL (mirror) = 4 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.591″ X 1.125 Y 1.625

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.500
P3	3.000	3.500
P4	3.000	5.000
P5	3.750	0.625

FREE-FORM CURVE 1 (grey dots, follow DXF) — points X, Y

3.065	4.997
3.130	4.989
3.194	4.974
3.256	4.955
3.317	4.930
3.375	4.899
3.430	4.864
3.482	4.824
3.530	4.780
3.575	4.732
3.614	4.680
3.650	4.625
3.680	4.567
3.705	4.506
3.724	4.444
3.739	4.380
3.747	4.315
3.750	4.250

FREE-FORM CURVE 2 (grey dots, follow DXF) — points X, Y

3.747	0.564
3.738	0.503
3.723	0.444
3.702	0.386
3.676	0.330
3.645	0.278
3.608	0.229
3.567	0.183
3.522	0.142
3.472	0.105
3.420	0.074
3.364	0.048
3.306	0.027
3.247	0.012
3.186	0.003
3.125	0.000

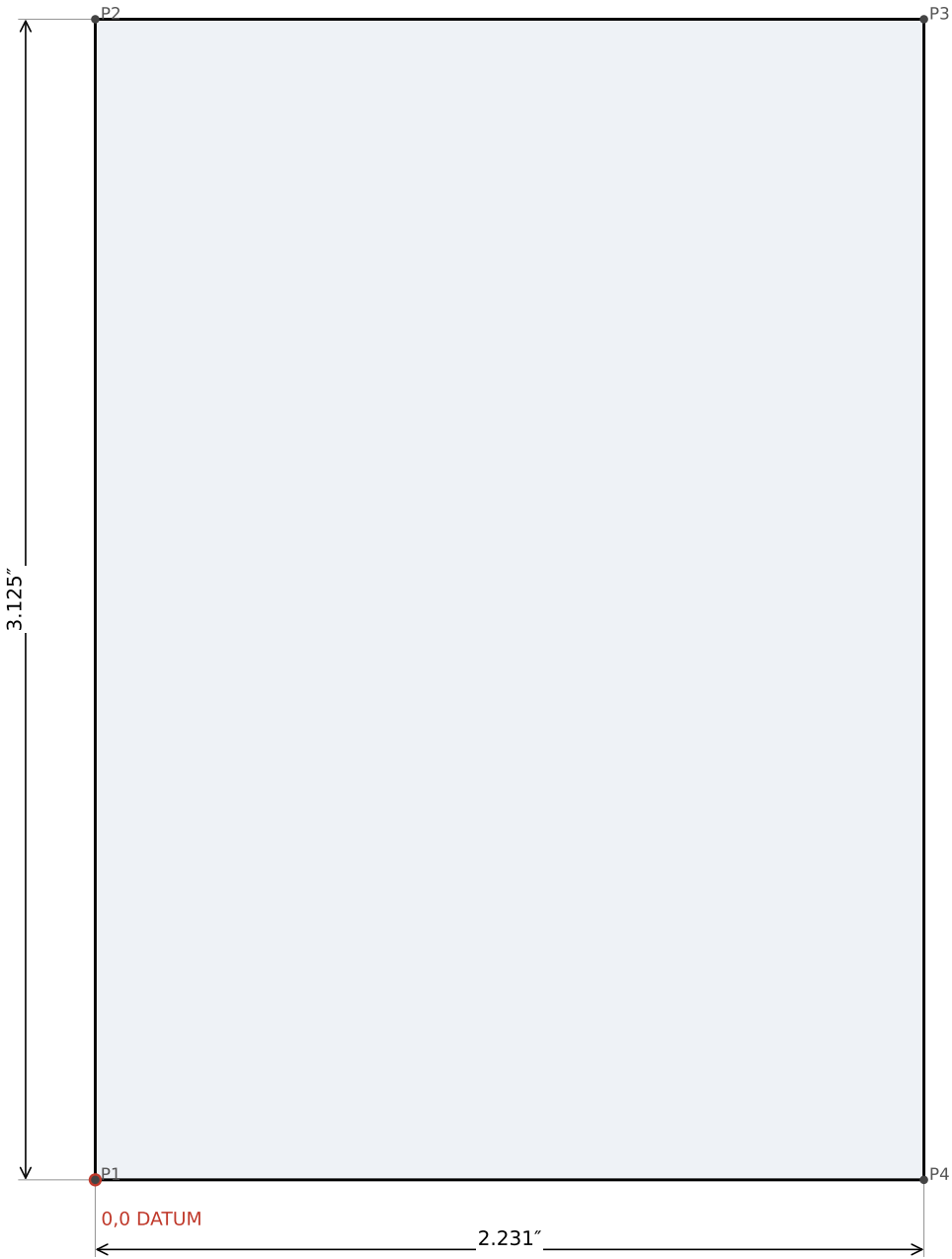
NOTES

1. Cut from the DXF file (1:1, inches). This sheet is for checking.
2. Material: HR steel sheet 10 ga (0.135″) (t). Inside bend radius = t.
3. Flat pattern already includes bend allowance (K = 0.44, R = t).
4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
5. Deburr all edges. Curved edges: smooth arcs per DXF.
6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 1 of 3	HR steel sheet 10 ga (0.135″)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate with tab	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010″ · bends ±1° · IR = t	SHEET 1 / 3

FR-7 (welded version) — blank 2 of 3

Bottom plate · blank 2.231" × 3.125" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	3.125
P3	2.231	3.125
P4	2.231	0.000

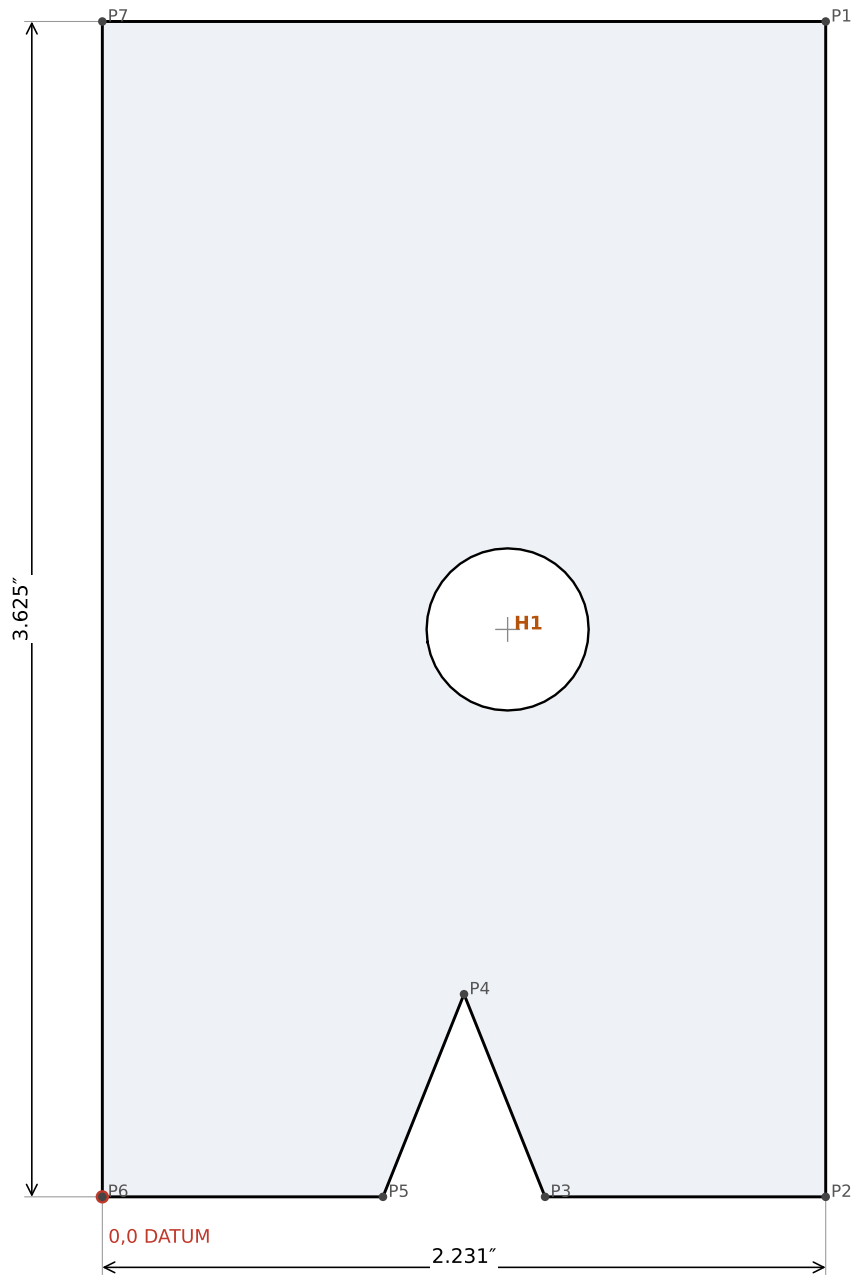
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 2 of 3	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Bottom plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 3

FR-7 (welded version) — blank 3 of 3

Back plate (hole Ø0.5) · blank 2.231" × 3.625" · HR steel sheet 10 ga (0.135") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

H1 Ø0.500" X 1.250 Y 1.750

OUTLINE POINTS (X, Y from datum)

P1	2.231	3.625
P2	2.231	0.000
P3	1.365	0.000
P4	1.115	0.625
P5	0.866	0.000
P6	0.000	0.000
P7	0.000	3.625

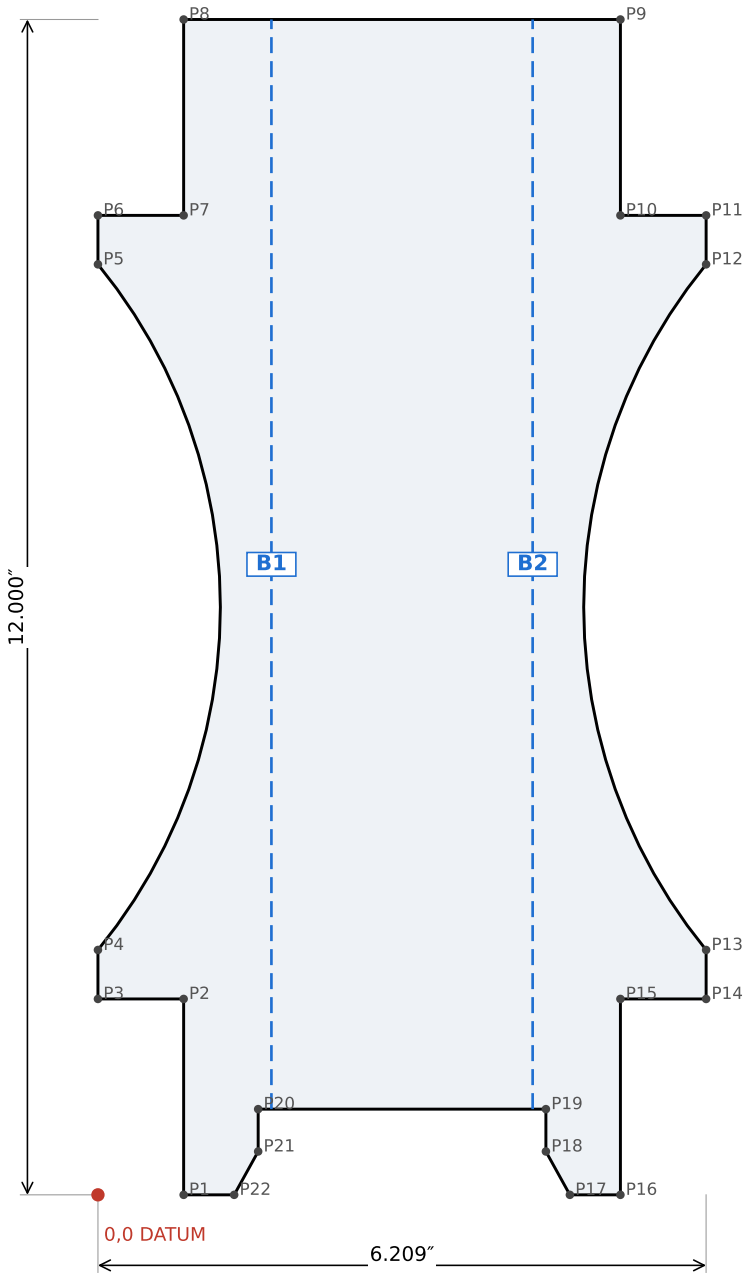
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 10 ga (0.135") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-7 (welded version) — blank 3 of 3	HR steel sheet 10 ga (0.135")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Back plate (hole Ø0.5)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 3 / 3

FR-8 (bent version) — blank 1 of 1

Channel (top + 2 sides) · blank 6.209" × 12.000" · HR steel sheet 11 ga (0.120") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

BENDS (bend centre line, from datum) IR = t = 11 ga (0.120")

B1 (1) 90° (1.771, 0.875) → (1.771, 12.000)
B2 (2) 90° (4.438, 0.875) → (4.438, 12.000)

CURVED EDGES (radius, centre X, Y)

R1 R 5.525" centre -4.275, 6.000
R2 R 5.525" centre 10.484, 6.000

OUTLINE POINTS (X, Y from datum)

P1 0.875 0.000
P2 0.875 2.000
P3 0.000 2.000
P4 0.000 2.500
P5 0.000 9.500
P6 0.000 10.000
P7 0.875 10.000
P8 0.875 12.000
P9 5.334 12.000
P10 5.334 10.000
P11 6.209 10.000
P12 6.209 9.500
P13 6.209 2.500
P14 6.209 2.000
P15 5.334 2.000
P16 5.334 0.000
P17 4.818 0.000
P18 4.574 0.443
P19 4.574 0.875
P20 1.636 0.875
P21 1.636 0.443
P22 1.391 0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
 - 2. Material: HR steel sheet 11 ga (0.120") (t). Inside bend radius = t.
 - 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
 - 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
 - 5. Deburr all edges. Curved edges: smooth arcs per DXF.
 - 6. FR side as drawn; FL side = mirror image (flip the blank).
- * 90° bends, R = t; relief cut at B

FR-8 (bent version) — blank 1 of 1	HR steel sheet 11 ga (0.120")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Channel (top + 2 sides)	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 1 / 1

FR-8 (welded version) — blank 1 of 2

Side plate with arc · blank 12.000” × 1.875” · HR steel sheet 11 ga (0.120”) · 2 per side · FR + FL (mirror) = 4 per vehicle

HOLES (centre X, Y from datum)

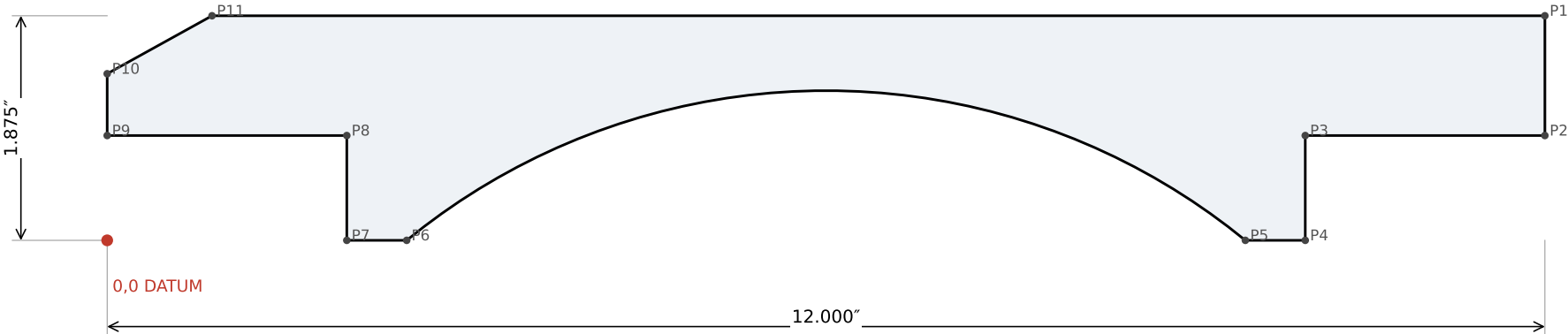
none

CURVED EDGES (radius, centre X, Y)

R1 R 5.525” centre 6.000, -4.275

OUTLINE POINTS (X, Y from datum)

P1	12.000	1.875
P2	12.000	0.875
P3	10.000	0.875
P4	10.000	0.000
P5	9.500	0.000
P6	2.500	0.000
P7	2.000	0.000
P8	2.000	0.875
P9	0.000	0.875
P10	0.000	1.391
P11	0.875	1.875



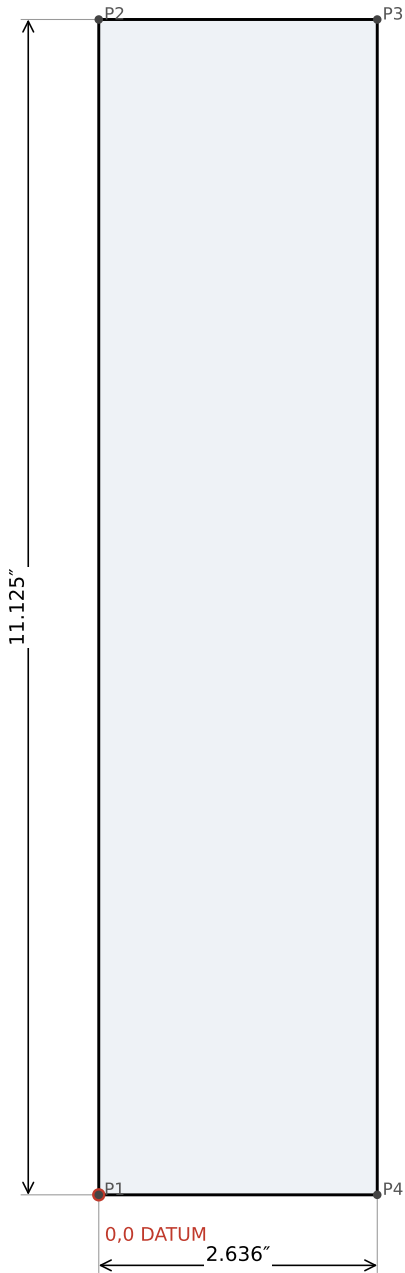
NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 11 ga (0.120”) (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-8 (welded version) — blank 1 of 2	HR steel sheet 11 ga (0.120”)	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Side plate with arc	QTY: 2 per side · FR + FL (mirror) = 4 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 2

FR-8 (welded version) — blank 2 of 2

Top plate · blank 2.636" × 11.125" · HR steel sheet 11 ga (0.120") · 1 per side · FR + FL (mirror) = 2 per vehicle



HOLES (centre X, Y from datum)

none

OUTLINE POINTS (X, Y from datum)

P1	0.000	0.000
P2	0.000	11.125
P3	2.636	11.125
P4	2.636	0.000

NOTES

- 1. Cut from the DXF file (1:1, inches). This sheet is for checking.
- 2. Material: HR steel sheet 11 ga (0.120") (t). Inside bend radius = t.
- 3. Flat pattern already includes bend allowance (K = 0.44, R = t).
- 4. Blue dashed = bend centre lines (B1, B2 ...). ↓ = bend DOWN (away from you), no arrow = bend UP. Red dashed = drill later.
- 5. Deburr all edges. Curved edges: smooth arcs per DXF.
- 6. FR side as drawn; FL side = mirror image (flip the blank).

FR-8 (welded version) — blank 2 of 2	HR steel sheet 11 ga (0.120")	inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Top plate	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010" · bends ±1° · IR = t	SHEET 2 / 2

FR-9 — cut list (tubes)

• Tube 3×3 — cut 8”

Stock: Square tube 3” × 3”, wall 3/16” · cut length 8.000” · QTY 1 per side · FR + FL (mirror) = 2 per vehicle

wall 3/16”

Cut square, ±1/32”. Deburr. Lengths in inches.

FR-9 — tubes		inches · flat blank, 1:1 in DXF	Sienna WAV frame · 2026-10-02
Tube 3×3 — cut 8”	QTY: 1 per side · FR + FL (mirror) = 2 per vehicle	cut ±0.010” · bends ±1° · IR = t	SHEET 1 / 1